GOVERNMENT
OF THE REPUBLIC OF MOLDOVA

MINISTRY OF ECONOMIC
DEVELOPMENT AND DIGITALIZATION
OF THE REPUBLIC OF MOLDOVA

UNDP

Republic of Moldova Digital Transformation Strategy

2023–2030

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List of abbreviations

AI	Artificial Intelligence
APS	Agency of Public Services
CDO	Chief Information Officer
CIO	Chief Data Officer
DESI	Digital Economy and Society Index
DPMD	Deputy Prime Minister for Digitalization
EGA	e-Government Agency
GDP	Gross Domestic Product
GNI	Gross National Income
IoT	Internet of Things
ITU	International Telecommunication Union
MTBF	Medium-Term Budgetary Framework
MDTS	Moldova Digital Transformation Strategy
ML	Machine Learning
MoE	Ministry of Economy
MoER	Ministry of Education and Research
MoF	Ministry of Finance
NARECIT	National Agency for Regulation in Electronic Communications and Information Technology
NBM	National Bank of Moldova
NBS	National Bureau of Statistics
NDTC	National Digital Transformation Council
R&D	Research and Development
RRP	Recovery and Resilience Plan
SME	Small and Medium-Sized Enterprise
ITSec	Information Technology and Cybersecurity Service
UI	User Interface
UX	User Experience
UNDESA	United Nations Department of Economic and Social Affairs
UNDP	United Nations Development Program
USAF	Universal Service and Access Fund

CHAPTER I

1. Introduction

The fourth industrial revolution is fundamentally changing global development and transforming society at an unprecedented pace. The COVID-19 pandemic has highlighted the reliance on digital tools and timely data and emphasized the need for digital transformation, which has become one of the highest priorities for societies, including Moldova. Digital transformation involves the integration of digital technologies in all aspects of a field of activity, which fundamentally changes the way it used to operate, provides distinct added value for stakeholders, and requires a new cultural orientation, focusing on continuously challenging the status quo and searching for the best solutions. The digital transformation is also characterized by customer-oriented organizational changes, supported by leadership, and driven by significant environmental and mindset challenges. Digital transformation causes or influences radical changes in all aspects of human life.

Following the conclusion of the national Digital Moldova strategy in 2020, approved by Government Decision no. f857/2013, Moldova needs to rethink its opportunities for digital transformation, using a whole-of-society approach, identify key opportunities to engage with people, business, local public administration, and build the demand for digital skills and digital solutions.

The Government Activity Program of Moldova, which identifies digital transformation as one of most important policy objectives for the next four years, is aligned with the European Union Association Agreement and the United Nations' 2030 Agenda for Sustainable Development. The recent acceptance of Moldova for European Union candidate membership (as of 23 June 2022) is another mobilization factor for integrating the country into the European Union legislation fields and the single digital market. In this context, it is necessary to accelerate already initiated transformational processes, align the national strategic measures with European and global trends and implement new policies based on the latest cross-cutting priorities of the European Union and the specific needs of the Republic of Moldova.

As recently attested by multiple international and local evaluation reports detailed further in this section, Moldova has continued to advance in building the main components for its digital economy and society. For instance, the Digital Readiness Assessment, developed by UNDP, recognizes Moldova as a country with all required prerequisites for a rapid and impactful digital transformation.

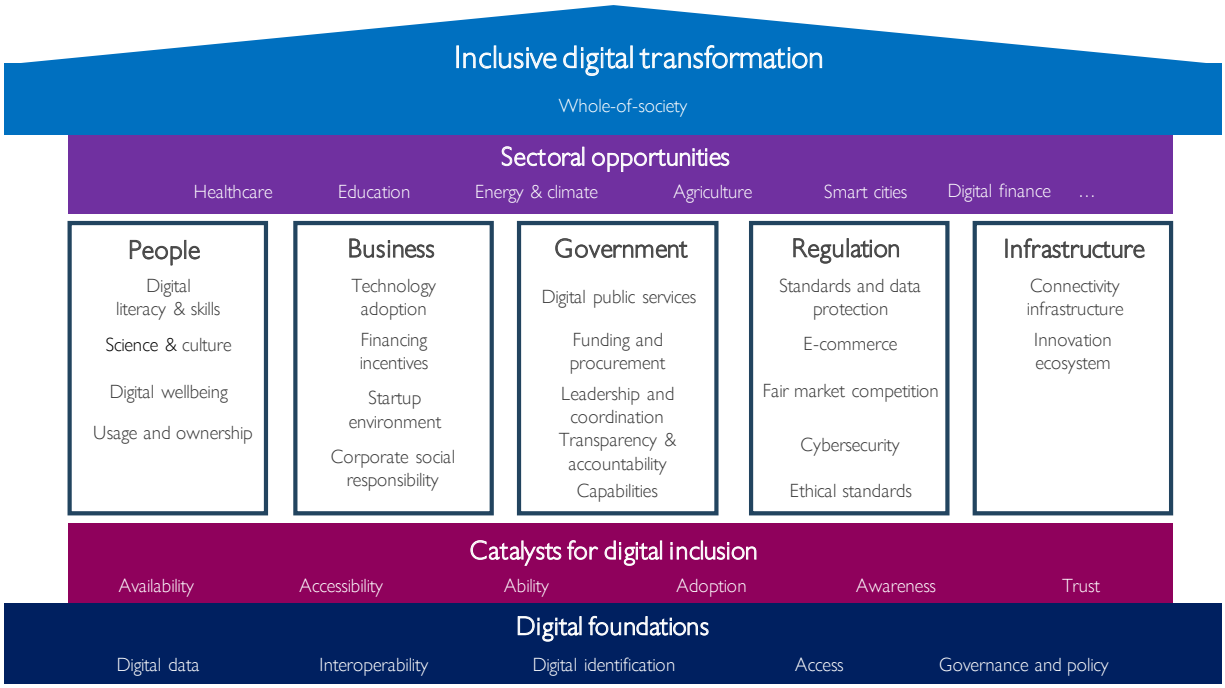
As such, the key pillars of a successful digital transformation for sustainable human development in the Republic of Moldova are:

- **people**, the most valuable resource, the catalyst for digital change, and the main beneficiary;
- **business**, the leader in digital transformation, technology adoption and development of innovative products and services in all sectors of the national economy;
- **government**, which creates the right conditions for the development of an inclusive digital society, develops enabling policies and regulations, and builds the supporting infrastructure for digital transformation;
- **regulations**, which provide an enabling legal and policy environment conducive to digital transformation;

- **infrastructure**, which represents the foundation of digital transformation and provides the main building blocks for any subsequent digital transformation reforms.

By focusing digital transformation efforts along these pillars, relying on, and further consolidating the digital foundations and making sure the catalysts for digital inclusion are in place, Moldova will be able achieve an inclusive digital transformation journey using an innovative whole-of-society approach.

Figure 1. Architecture of the Moldova Digital Transformation Strategy 2022-2030



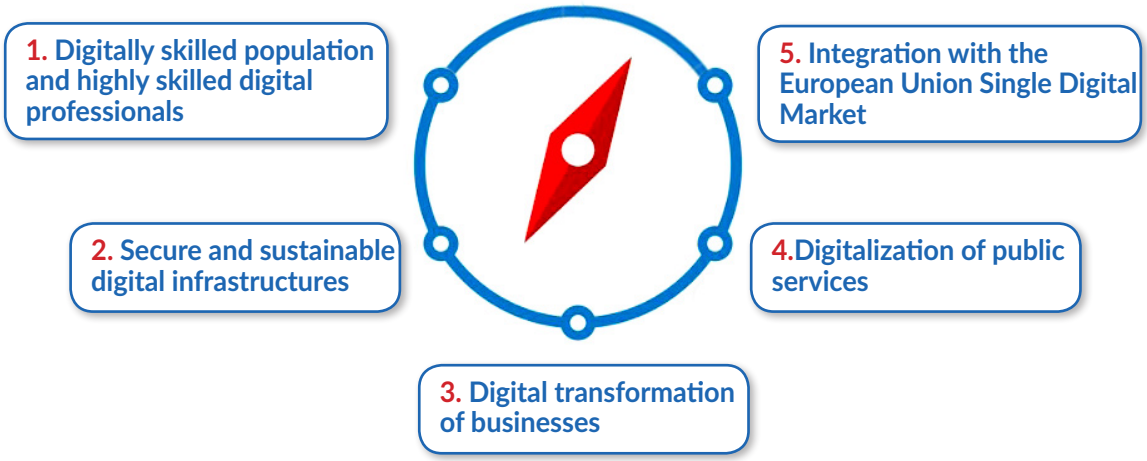
Source: developed based on the Assessment Analysis of the degree of readiness for digital transformation of Moldova, UNDP

The results of the extensive consultations carried out in the process of preparing the strategy highlight that the stakeholders (central public authorities, local public authorities, business, academia, civil society, etc.) regard the development of a new national strategy as an important and timely tool for guiding Moldova's digital transformation efforts.

The main areas of intervention will therefore follow the four cardinal points from the European Union's Digital Compass (https://digital-strategy.ec.europa.eu/en/policies/europes-digital-decade#ecl-inpage-kyvdstob) with the addition of across-cutting dimension for integration within the European Union Single Digital Market:

1. Digitally skilled population and highly skilled digital professionals
2. Secure and sustainable digital infrastructures
3. Digital transformation of businesses
4. Digitalization of public services
5. Integration with the European Union Single Digital Market

Figure 2. Main intervention areas of the MDTS



Source: adapted by the authors of the European Union's Digital Compass

The Moldova Digital Transformation Strategy (MDTS) is thus a citizen-centric document aimed at improving people's well-being, with a basic mission to transform the country's economy and society with digital tools.

To support the digital transformation of the economy and society in Moldova, the authors involved in developing the strategy combined strategic thinking and a long-term vision, allowing the strategy sufficient flexibility to adjust to the fast-changing pace of the digital environment to increase its likelihood of achieving established targets.

A robust governance and coordination structure with clear roles, responsibilities, and comprehensive monitoring and reporting mechanisms, with inclusive follow-up and efficient risk management, is proposed in the document to ensure smooth implementation and to achieve the targeted goals. The structure will be created under the auspices of the Ministry of Economic Development and Digitization, as the institution responsible for development of policies, and the agencies and institutions responsible for the implementation of policies in the targeted areas. The governance structure will also include the National Digital Transformation Council (NDTC), the Cyber Security Coordinating Council, the Economic Council under the Prime Minister of Moldova and the implementing strategic units.

The monitoring and evaluation framework will be based on the Digital Economy and Society Index (DESI) by adapting the DESI Methodology for Republic of Moldova, as well as including other relevant indicators. Such a framework will allow objective evaluation of implementation progress, of investment efficiency and identification of the potential for improvement.

The strategy will serve as a guiding and orientation document for the central and local public authorities, the business community, academia, civil society, strategic development partners and for targeting, planning, financing, implementing, and monitoring digital transformation agendas until 2030.

By 2030, due to implementation of the strategy, Moldova will have an innovative and inclusive digital society with digital competences based on a modern digital infrastructure, with digital governance and a business community that makes full use of digital opportunities.

CHAPTER II

ANALYSIS OF THE SITUATION

SECTION 1. Digital environment in the Republic of Moldova

The pandemic turned digital transformation from a priority into a global imperative, and this was also true for the Republic of Moldova. Over the past two years, the COVID-19 pandemic has increased dependence on technology and accelerated the inevitable process of digital transformation as many aspects of life have moved online. The shift towards improved technological capability and integration presents an opportunity to build a better and more connected future considering the "new normal".

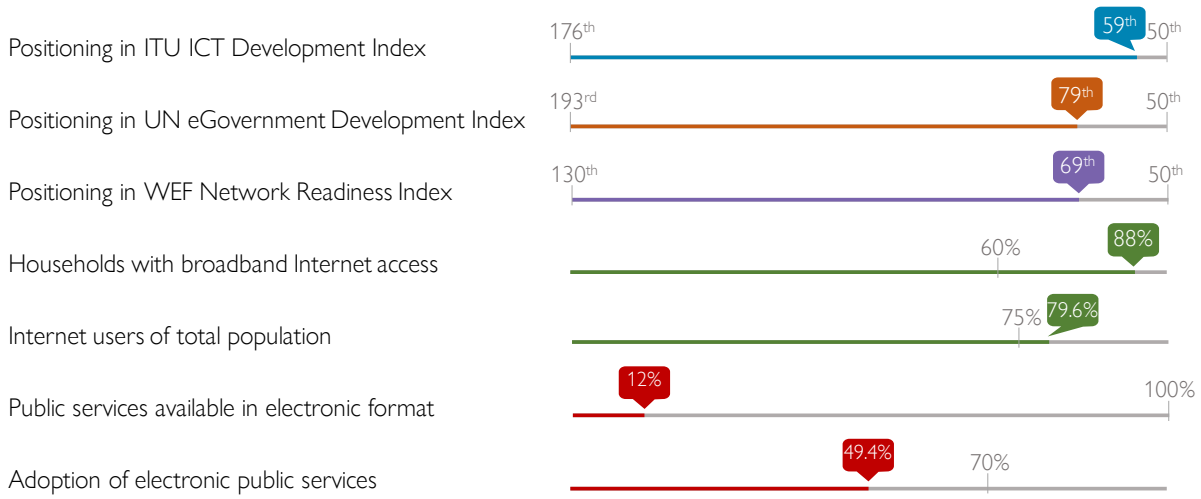
Over the last two decades, the digital transformation efforts of the Republic of Moldova have been guided by several national and sectoral strategies, such as the National Strategy of Information Society Building 'Electronic Moldova' approved by Government Decision no. 255/2005, the National Strategy for Information Society Development 'Digital Moldova' 2020 approved by Government Decision no. 857/2013, the Strategic Program for Technological Modernization of Government (e-Transformation) approved by Government Decision no. 710/2011, the Strategy for the IT Industry and Digital Innovation Ecosystem Development 2018–2023, approved by Government Decision no. 904/2018, the Information Security Strategy of the Republic of Moldova 2019–2024, the National Research and Innovation Program 2020–2023 approved by Government Decision no. 381/2019 and the Radio Spectrum Management Program 2021–2025 and the minimum exposure values of generic frequency blocks for allocation through the competitive selection procedure approved by Government Decision no. 987/2020.

The current Government Activity Program includes digitalization among the 15 sectoral priorities. The National Development Strategy 'European Moldova 2030' approved by Law no. 315/2022, aligned with both the European Union Association Agreement and the United Nations 2030 Agenda for Sustainable Development, dedicates a special Chapter to digital transformation of the government, company and economy.

However, successful implementation of strategic documents depends on many factors and challenges. As outlined in the final report on the implementation of the Digital Moldova 2020 action plan, there have been positive developments in terms of creating conditions for the implementation and use of electronic services, the development of digital skills, improving connectivity and network access and eliminating critical constraints on the information and communication technology (ICT) business environment. At the same time, the report notes certain failures caused mainly by insufficient funding and lack of qualified human resources, implementation delays and inaction.

Those deficiencies were caused by the limited institutional capacities for the development, implementation and coordination of ICT projects in the public sector and insufficient resources for maintenance and development.

Figure 3. Results indicators for the national Digital Moldova 2020 strategy (target versus actual)



Source: developed by the authorss

As can be seen from Figure 3, the specific objectives related to e-government have not been fully achieved, although there have been remarkable developments regarding the e-transformation of government, especially the growth of the ICT sector.

Thus, on 31 March 2023, out of the total (517) public administrative services (G2B) registered at eGA, 204 services are available online. Out of 473 public services available G2C, 154 services are provided online.

The ICT sector has become the main driver of digitalization and innovation in Moldova and is growing fast. In 2021, the IT industry reached a share of more than 4.25 per cent of gross domestic product (GDP), exceeding 10 billion MDL of sales, with the share of the ICT sector being over 7.6 per cent of GDP with other 18 billion MDL of sales in 2021 reached by about 2000 companies with other 30.000 employees. According to the NBM report for 2022, the export of ICT services reached USD 501.85 million IT exports are growing at an annual rate of over 30 percent over the past 5 years and reached a record high level of USD 468.67 million.

The single 7 per cent sales income tax offered by the Moldova IT Parkand the wide range of eligible activities, including ICT research and development (R&D), have given a significant impetus to the development of the ICT sector. In its five years of activity, the Moldova IT Park has become one of the most successful IT sector development initiatives in Moldova, having attracted over 1395 residents, with about 18,700 employees (of whom: men 68%, women 32%) and a forecasted revenue for 2023 of over 10.7 billion MDL.

According to the Speed test Global Index, (https://www.speedtest.net/global-index) which provides a monthly comparison of Internet speed data for a benchmark of 100 countries around the world, the Republic of Moldova is ranked fifty-eighth in terms of mobile broadband speed (with a download speed of 38 Mbps compared to the global average of 48 Mbps), and is ranked thirty-eighth in terms of fixed broadband speed (with a download speed of 106 Mbps – higher than the global average of 98 Mbps). Moldova is

among the top 10 countries in the world in terms of accessibility and affordability of Gigabit Internet access, where the users have access to unlimited Gigabit Internet for about EUR 15 per month.

The United Nations Department of Economic and Social Affairs e-Government Index (https://www.un.org/development/desa/publications/publication/2020-united-nations-e-government-survey) attests to this considerable access to the Internet and to IT devices in the Republic of Moldova, although existing electronic communications infrastructure deployment is below the Eastern Europe subregion average, requiring further efforts to keep the country competitive.

Market demand and available content are key elements for network development. With 98 per cent of the population covered by a 4G signal and similar fiberoptic backbone network penetration, users have the freedom to choose their preferred means of connectivity. Internet access is relatively affordable, and its use is widespread, with 80 per cent of population accessing Internet services in 2021. However, according to the data collected within the Generations and Gender Survey 2020 carried out by UNFPA, the digital gap between the elderly and young people in Moldova is huge – only 34% of the population aged between 60 and 79 used the Internet, compared to 82% of those aged 15-59. The share of women who use the Internet is 2.1% higher than the share of men, and represent 74.7% and 72.6% accordingly.

Government services have seen rapid digitalization rates with the establishment of the eGovernment Center (now the E-Government Agency) in 2010. A range of reusable digital platforms have been developed and successfully implemented, including the common government technology platform (MCloud), the government electronic authentication and access control service (MPass), the government electronic payment service (MPay), the interoperability platform (MConnect), Government Open Data Portal, etc.The public services portal,which has an average of over 0.5 million unique visitors per year, is the one-stop shop for all public services, offeringsingle-sign-on access to 166 eservices and complete and concise information about another 531 public services. The portal functions as an electronic catalogue of public services for citizens and businesses and is aligned with the European Union Single Digital Gateway, following the Core Public Service Vocabulary data model (https://joinup.ec.europa.eu/collection/semantic-interoperability-community-semic/solution/core-public-service-vocabulary).

Several documents produced by international development partners and national stakeholders in 2021 have provided a detailed analysis of the current situation on ICT, innovation and e-read inessand highlighted the main gaps and opportunities for the digital transformation of Moldova. These documents paint an accurate picture of the digital development barriers and opportunities in the country and have made a valuable contribution to the development of the current strategy.

The main problems identified in the reports mentioned in the reference, as well as during the strategy consultation and interview processes, include:

- Low R&D expenditure, slow growth on innovative companies, lack of venture capital and low university–industry cooperation;

- Lack of legal acts on the new digital technologies (such as Artificial Intelligence (AI), blockchain, Internet of Things (IoT), Big Data and mining) and insufficient adaptivity of the legal framework to digital business models;
- Mismatch of digital skills, as well as an insufficiently skilled workforce in the industry and central public authorities, especially the local public authorities;
- Low level of digital skills and awareness among the population;
- Emerging level of digital readiness of Moldova's education system (level 2 of 4) (https://documents1.worldbank.org/curated/en/099120006252220689/pdf/P17773104ea6f2040a88e02bdf9bbd04f6.pdf);
- Insufficient institutional capacities for the implementation of the digital transformation agenda;
- Reluctance to digital transformation within service providers;
- Shortage of financial resources;
- Underdeveloped IT infrastructure in local public authorities;
- Low level of ICT investment in the agricultural small and medium-sized enterprises (SMEs);
- Low involvement of local companies in government-financed IT projects;
- Insufficient remuneration of public servants responsible for managing the ICT infrastructure and digitization projects of the institution;
- Insufficient presence of scientific research in the development and monitoring of policies aimed at developing the information society.

Nevertheless, the Moldova Digital Readiness Evaluation Analysis acknowledges that the country possesses the essential elements for digital transformation, while also identifying certain weaknesses that need addressing.

Infrastructure

- A well-developed mobile infrastructure offering strong potential for the ICT market is present in the country, but highspeed home access to broadband is still challenging.
- Affordable mobile Internet access exists, but the fixed broadband basket cost does not yet meet the Commission's target.
- There is strong government support for ICT sector development, but no joint vision or overarching strategy related to its further development.
- There have been a few efforts by stakeholders to increase cyber-resilience, but there is a clear need for active engagement in strengthening the country's cyber defense capabilities.

People

- The capacity of citizens to develop technical capabilities is expanding but still does not meet the level of demand, and there do not seem to be rewards or incentives to encourage entrepreneurial mindsets or further societal adoption
- IT professionals in Moldova have good qualifications, but the size of the talent pool is very limited and not integrated; a mismatch of skills exists
- Significant efforts are undertaken by the government and other stakeholders to ensure an enabling environment for the digital inclusion of women and girls, but barriers remain for career advancement; programs do not target the inclusivity of the most vulnerable groups

- Moldova has a solid mandatory education system that lays the foundation for advanced professional digital skills, yet there are still too few programs which can attract sufficient talent into pursuing STEM careers and address the lack of specialized IT skills among the population
- National stakeholders engage actively in promoting online protection, but there is a need for a more robust and coordinated approach
- Even with wide Internet access and ownership of ICT devices, challenges can be identified in the degree of quality and affordability of access for certain vulnerable groups (women, rural areas, the elderly) that limit their capacity to access some services and hamper comprehensive development of the digital economy

Government

- eGovernment development is improving, with many laws and frameworks being developed, but ranking on eGovernment index is still low
- Public sector transformation is gradually enabling effortless digital interaction between government authorities and citizens, and strives to provide high-quality public services
- Ongoing public administration reform embraces coordination with local and national stakeholders, data sharing and cooperation between institutions
- Several relevant online services are available for citizens on key areas related to governance and transparent procurement; however, many essential services to citizens are yet to be digitized, and further efforts are needed. The widespread use of digital ID, for instance, isone of the key enablers of more digital services being developed and more people using the digital services
- There is not much awareness regarding the use of open-source technologies around government procurement, indicating a gap and potential barrier in terms of the opportunity to expand the local ICT sector
- Lack of coordination between governmental agencies and of a concrete vision reverts to technical problems in system architectures, interoperability, and service integration. Further work should be undertaken on customer orientation, monitoring and evaluation mechanisms and addressing internal change resistance and bureaucratic barriers to deploy new services 4)

Business

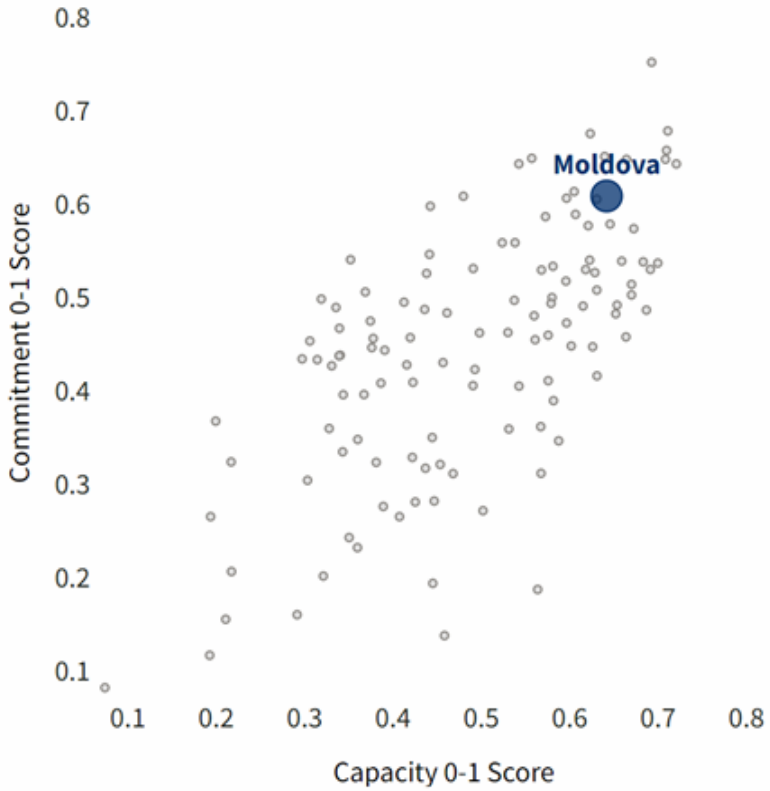
- SMEs are the backbone of the economy, with huge untapped potential, but with an urgent need to digitally transform their businesses
- Entrepreneurs can easily create a business in Moldova and the country ranks very positively in the domain of ease for creating a business, but this contrasts with the lack of credit and financing for the private sector and for new ventures, which is limited in the country and discourages the emergence of new start-ups and business models
- The start-up sector is a key source of innovation, but few initiatives are in place for growing up the newly created businesses
- The ICT sector is, generally, a driving for cefor development and digital transformation, but private enterprises benefit too little of the advantages it offers
- The agricultural and food production industry is a priority sector for digital transformation, but there is a low level of readiness of this and other related sectors, such as transportation

Whole-of-society

- The performance of Moldova in key indicators of ICTs, innovation and entrepreneurship should translate into its competitiveness but there is still significant space for improvement
- There are widespread efforts at policy reforms, but there is no common vision for innovation ecosystem development engaging all the stakeholders
- The soft infrastructure is well developed in the capital city and two new infrastructure elements are under development in the regions, yet a big part of the country remains underserved
- The ecosystem lacks sustainable funding and talent readiness to ensure digital innovation at scale

According to the Moldova Digital Readiness Assessment (https://www.undp.org/moldova/publications/digital-readiness-assessment), Moldova scored 4.1 points out of a maximum of 5, putting it at the Differentiating Stage – the fourth of five stages of digital transformation readiness. This means the country has clear strengths in digital transformation capacities and the foundational elements are in place. Furthermore, USAID's FY2022 Moldova Country Roadmap (https://roadmaps.usaid.gov/country/moldova) indicators for commitment and capacity were at 0.61 and 0.64, which indicates that Moldova has the capacity to fulfil the assumed commitments, including the commitments included in the MDTS (Figure 4). These trends are also reflected in the World Bank GovTech Maturity Index study, according to which in the last two years Moldova surpassed the maturity index of countries with a significant focus on GovTech and entered the group of leading GovTech countries (https://www.worldbank.org/en/programs/govtech/gtmi).

Figure 4. The overall Commitment and Capacity scores for low- and middle-income countries, Moldova FY2022 roadmap, USAID



SECTION 2. Global and European Union trends

Current trends regarding the transition from industrial society to information society and Industry 4.0 constitute a revolutionary global change comparable to the arrival of the steam engine in the nineteenth century, of electrical energy in the first half of the twentieth century, and of the Internet at the end of the same century. The main pillars of Industry 4.0 are cybersecurity, cloud computing, mobile technologies, machine-to-machine, 3d printing, advanced robotics, Big Data/analytics, IoT, RFID technologies and cognitive computing.

Digital transformation is currently one of the most important approaches in the process of building and managing strategies and visions for countries, companies, and organizations. Based on the latest studies and analyses of global scientific and analytical companies and expert media, (https://www.forbes.com/sites/danielnewman/2017/09/26/top-10-trends-for-digital-transformation-in018/#2bd2d1c293ab) the most important technologies from the global viewpoint are AI; IoT; 5G technology; BigData and analytical data processing; Edge computing and cloud computing; blockchain; quantum computing;-datafication; machine learning; extended reality; Digital Trust; 3D printing; genomics; and new energy solutions.

The 2030 Agenda for Sustainable Development, adopted by Member States of the United Nations Organization in 2015, is a summary of global commitments through which the international community is calling for a joint coordinated procedure for solving the most serious global challenges, which include climate change, poverty, increasing economic and social inequalities and the unsustainability of prevailing production and consumption patterns (https://sustainabledevelopment.un.org/content/documents/21252030%20Agenda%20for%20Sustainable%20Development%20web.pdf).

In Moldova, the 2030 Agenda is used to determine long-term development priorities for the development of the country, as reflected in the National Development Strategy 2030. Specifying development priorities is directly connected to the identification of goals for the digital development of Moldova. At the national level, there are seven specific objectives reflecting the specific situation in Moldova, which may serve as the basis for future strategic and conceptual work, including their incorporation into the vision and priorities of this strategy. These are:

1. Developing opportunities for innovation and entrepreneurship
2. Ensuring universal access to the Internet and electronic services
3. Training for relevant skills in vocational and higher education
4. Developing a flexible and relevant system of continuing adult education
5. Focusing public services on people's needs
6. Integrating science, technology, and data into the governance process
7. Increasing security in the operation of critical infrastructures and systems

The digital and AI strategies of countries that are advanced in digital transformation prevailingly contain the following components:

- Support to R&D in new technologies and AI
- Support to innovative SMEs, which can be applied in practice either by means of know-how and incubators or, more particularly, thanks to venture capital

- Rules and standards, e.g., for creating, sharing, and managing data or the IoT
- Simplification, acceleration, and release of regulations to enable innovations and experiments, as well as to review impacts of the platform economy and labor changes in the digital era on the labor law and its institutes in the context of entities affected by it
- Development of talent that can design and use digital innovations, such as through a sophisticated migration policy as well as by creating interesting conditions for life in key towns
- Building digital infrastructure, e.g., public electronic services and sectoral platforms
- Building new models of cooperation among industry, academia, and the government sector

The Digital Europe Program represents an investment framework of the European strategy for the Digital Single Market, which responds to the trends of the digital era. With proposed investments of more than EUR 9.2 billion, the Digital Europe Program supports the digital transformation of European society and economy (https://eur-lex.europa.eu/legal content/EN/TXT/?uri=COM%3A2018%3A434%3AFIN).

The program has five specific goals to which the proposed budget will be allocated:

1. Supercomputers/high-performance computing (EUR2.7 billion)
2. AI (EUR 2.5 billion)
3. Cybersecurity and trust (EUR 2 billion)
4. Advance-d digital skills (EUR 0.7 billion)
5. Digital transformation and interoperability (EUR 1.3 billion)

The Digital Europe Program will strengthen capacities in Europe in key areas of digital technology and will support their dissemination and use in areas of public interest and in the private sector. The goal of the program is, inter alia, to adjust the European Union budget to future programming challenges, while the digital transformation is reflected in all proposals, ranging from transport, the energy sector and agriculture up to health care and culture.

The European Union Recovery and Resilience Facility represents an unprecedented opportunity for Member States to invest in their own digital transformation and to collectively contribute to increasing the European Union's resilience and innovative potential, as well as to reduce the European Union's external dependencies (https://ec.europa.eu/info/business-economy-euro/recovery-coronavirus/recovery-and-resilience-facility_en).

Under the Regulation (EU) 2021/241 of the European Parliament and of the Council of 12 February 2021 establishing the Recovery and Resilience Facility, each Member State must dedicate at least 20 per cent of its Recovery and Resilience Plan (RRP)'s total allocation to measures contributing to the digital transition or to addressing the challenges resulting from it. To date, an amount of EUR 127 billion, representing 26 per cent of the total allocation of the approved plans, has supported the digital transformation.

European Union's Digital Decade: digital target for 2030 (Digital Compass) presents a vision and avenues for Europe's digital transformation by 2030 and focuses efforts around four core areas https://ec.europa.eu/info/strategy/priorities-2019-2024/europe-fit-digital-age/europes-digital-decade-digital-targets-2030_en.

1. **Digitally skilled citizens**: The aim of the European Commission is that, by 2030, at least 80 per cent of all adults will have basic digital skills, and 20 million ICT experts will be employed in the European Union with a trend towards gender balance.
2. **Digitization of public services**: European Union citizens will be given the opportunity to deal online with all important official business and to retrieve their medical records electronically. By then, 80 per cent of the European Union population will also be able to use a digital ID.
3. **Digital transformation of businesses**: The plan is that, by 2030, 75 per cent of European Union businesses will use the Cloud, AI, and Big Data, and at least 90 per cent of SMEs will have a basic level of "digital intensity".
4. **Secure and sustainable digital infrastructures**: By 2030, all households in the European Union should have a gigabit connection, and all populated areas should have 5G coverage. A total of 10,000 high security and climate neutral datacenters should be in use, and Europe should have its first quantum computer. Twenty per cent of the world's cutting-edge and sustainable semiconductors should be produced in Europe.

To achieve these ambitious goals, "multi-country projects" will be launched as soon as possible. The Member States committed to allocate at least 20 per cent of expenditure to their recovery and resilience plans for digital transformation.

After obtaining the status of candidate country for EU accession, in addition to the implementation of the Association Agreement, the Republic of Moldova shall, among other actions, increase the capacity to implement reforms and provide quality public services, which cannot be ensured without the digital transformation of the country. This transformation can be carried out on the basis of a strategy aligned with the relevant documents of the European Union in the fields of digital society and economy, digital governance and cyber security.

CHAPTER III

MISSION, VISION AND OBJECTIVES OF THE STRATEGY

The Strategy lays the foundation for building a digital society in its entirety in which adoption and integration of ICT technology is ubiquitous, be it at home, at work, in education, in health, cultural and recreational contexts, etc.

To achieve this, the document will touch upon the existing and required prerequisites, enablers, and key principles of people-centric digital service design to meet the user expectations, standards of interoperability, data security and confidentiality.

The goal is not only to transform services in the public and private sectors for their use in the digital space, but to rethink and redesign them by reducing the need to fill in application forms, eliminating the need to submit information that is already available, and diversifying delivery channels. The strategy also addresses aspects related to creating and maintaining a trusted cyberspace, which will contribute to widespread adoption of technologies and to improving the socioeconomic development of Moldova, so that all the people can enjoy the full benefits of a secure, friendly, and resilient digital space.

SECTION 1. Mission of the strategy

The mission of the strategy is to help achieve effective public governance, competitiveness in all spheres of life and the well-being of people, enabling Moldova to become an European UnionMember State.

SECTION 2. Vision of the strategy

As a result of implementing the strategy, the Republic of Moldova will become an innovative, digitally-skilled and inclusive digital society, with an advanced digital infrastructure, pro-digital governance and a business community that makes full use of digital opportunities.

SECTION 3. General objectives

Based on the European and global trends and taking into consideration the digital readiness of Moldova for digital transformation, as well as the recommendations of the available diagnostic and evaluation reports and the results of the public consultation process, the MDTS will focus on six general objectives that will guide the activities under the strategy:

Based on the European and global trends and taking into consideration the digital readiness of Moldova for digital transformation, as well as the recommendations of the available diagnostic and evaluation reports and the results of the public consultation process, the

MDTS will focus on six general objectives that will guide the activities under the strategy:

1) Develop a digital society
2) Grow a robust and competitive ICT environment
3) Create an innovative and resilient digital economy
4) Establish an efficient, smart, and transparent digital state
5) Create a secure digital accessible and inclusive environment
6) Make Moldova a trusted and reliable digital nation

Achieving the above objectives will ensure a functional and secure environment for the widespread development and use of digital solutions in every sphere, including the removal of inequities and disparities associated with gender, living environment, disability and income, facilitating the European Union integration process.

SECTION 4. Core principles of strategy implementation

Many principles have laid the foundation for existing strategic documents in the field of digital transformation that are equally valid for the implementation of the current strategy. The nine Principles for Digital Development, (https://digitalprinciples.org/) for instance, have been endorsed by a large proportion of international stakeholders and provide practical guidance for digital transformation. However, there are three overarching principles the strategy is putting at the forefront and which are defining for the mission, vision and objectives set above.

1) **Human development over technology**: Digital transformation is first and foremost about putting people at the center, rather than focusing on the technological aspects of implementation. Technology is but a tool for achieving the vision and the objectives of the strategy whereas properly motivated and trained people are it score.
2) **Shared implementation responsibility** between central and local government, private sector, development partners and the entire society. Experience of previous strategies has shown that involving only a few stakeholders in implementation will likely result in siloed solutions, with stakeholders trying to solve complex problems from narrow institutional perspectives.
3) **Compliance with European Union standards and regulations** and reliance on tried and tested international best practices. The new country strategy of European Union integration dictates the need to harmonize the existing and future institutional, regulatory and technological framework with that of the EU by taking over common standards and best practices, including those recognized internationally. The creation of a joint Moldova–Romania common digital space would be an excellent test in this sense for ensuring compliance with European Union standards and regulations, given the strong cultural and social affinities between the two countries

CHAPTER IV

PRIORITY DIRECTIONS AND THEIR IMPACT

SECTION 1. Strategy implementation considerations

Based on the experience and lessons learned from the implementation of previous digital transformation strategies, this section will outline a few implementation considerations to help increase the likelihood of success of digital transformation activities under the current document. At the same time, given the long implementation time frame of this strategy and the need for constant adaptation and change, the strategy will be implemented through the following programs focused on achieving the general objectives:

1) Digital Society Development Program
2) ICT Sector and Digital Economy Development Program
3) Public Sector e-Transformation Program.

During strategy implementation, developing and approving implementation plans will help translate strategic directions into specific activities and projects, while allowing the government to introduce necessary correctives and leverage various funding streams as they become available.

At the same time, the implementation of the Strategy will be based on the following main aspects:

1) **Balancing quick wins and strategic activities** during strategy implementation: This is essential to provide early successes and increase the implementation momentum. The high expectations linked to development and approval of the strategy will need to translate into short-term gains that can reassures take holders about the feasibility of implementation, while providing the main implementation partners with sufficient time to deliver on strategic, heavy-lifting activities.
2) **Establishment of the NDTC**: Resistance to change, institutional inertia and departmental interests can significantly impact implementation of the activities under the strategy. Establishing the NDTC, will serve as an efficient mechanism to remove any major administrative and institutional barriers for digital transformation in a timely manner.
3) **Creation of the Strategic Delivery Unit for implementation of key activities**: Implementation of such an all-encompassing, transformative strategy can be overwhelming for many implementation actors; it may, at times, exceed their ability deliver complex projects (*such as health care*) and may lead to delays and failures. Creation of a dedicated Strategic Delivery Unit within thee-Government Agency, whose role would be to oversee implementation of key digital transformation programs and projects, will ensure that selected strategic activities and project will be better managed and delivered.
4) **Reliance on agile delivery models**: Recent experience of using agile methodologies in the public sector, including by the EGA, has clearly shown that agile delivery methods, while reusing existing platforms an doff-the-shelf services, will reduce costs and delivery times and result in products that are more adapted to user needs.

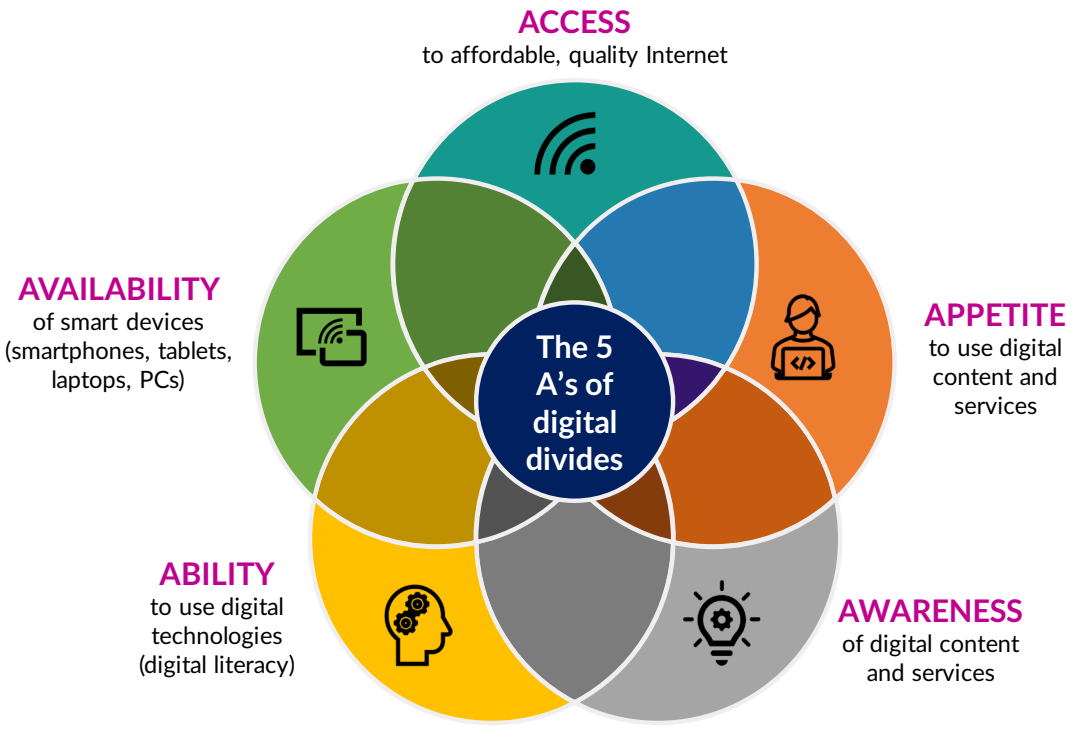
5) **Involving private sector companies** as key implementation partners of the strategy: This includes public procurements for government information systems and, where possible, opening APIs for developing innovative add-ons and solutions. This will contribute to market competition and a strong local ICT industry.

SECTION 2. General objective no. 1. Development of a digital society

Given the strong human centricity of the strategy, the first objective is to develop a digital society able to reap the benefits of digital technologies and services.

To assure widespread use of digital technologies and services in every aspect of life (social, economic, and political), every citizen of the country should know of them, have access to them and possess the ability to use them freely. By deep-diving into the building blocks of a digital society, it becomes clear that the primary dividing lines between those who have been able to adopt digital technologies in their lives, and those who have not, can be described using the five A's: access to affordable, quality Internet; availability of smart devices (possession/ownership); ability to use digital technologies (literacy and skills); a wareness of digital content, services and technologies; and their adoption, or the appetite to use them(relevant content/interest).

Figure 5. The 5 A's of digital divides



Source: adapted by the authors based on the European Union's Digital Compass

The Internet coverage, quality, afford ability and penetration indicators for Moldova are good for its region and income class. However, exclusions and gaps still exist – 34.3 per cent of those who are not connected invoked high prices as the main reason for being unconnected, while another 48.1 per cent and 20.3 per cent respectively lack computers and/or smartphones (https://old.egov.md/ro/resources/polls/raportul-sondajului-national-anual-2021).

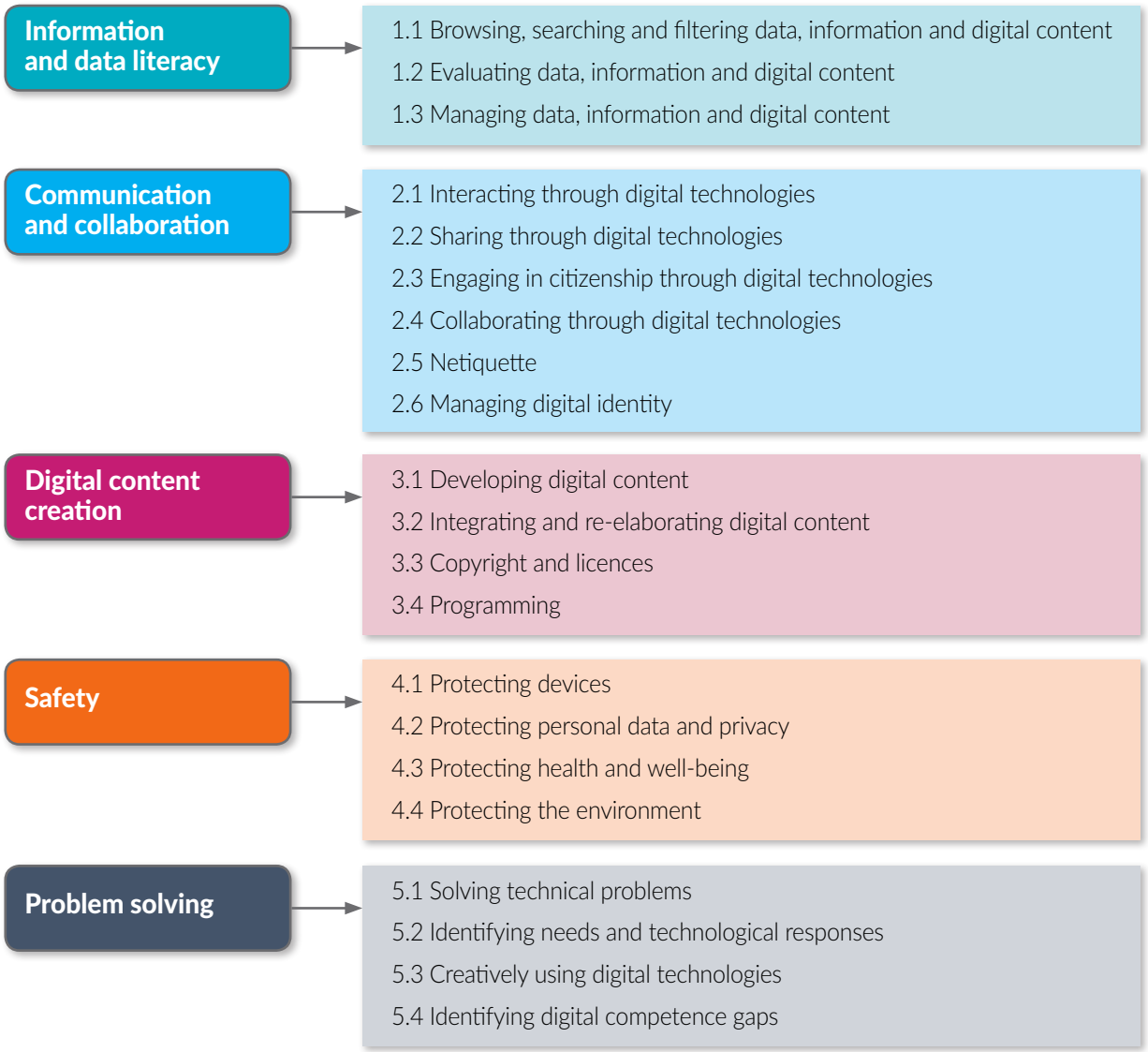
These values vary considerably across respondents' income levels and much less on their residence (urban/rural), gender, age and education.

With regards to ability and appetite, the EGA (https://old.egov.md/ro/resources/polls/raportul-sondajului-national-anual-2021) notes that only one in two respondents who accessed a public service has used an electronic service, with a huge gap of 60 per cent in educational levels and a 20 per cent gap between urban and rural population. Young people aged 18-29 are the category that almost five times more often used electronic services compared to people aged 60-74. The level of the indicator among men is about 10 percent higher than its level recorded among women.

Thus, a strong focus under this objective should be to avoid exclusion and reduce digital divides as much as possible. By some estimates, (https://a4ai.org/report/2021-affordability-report/) the digital gender gap alone cost the global economy over US$ 1 trillion. No one should be excluded by lack of access, skills, social and cultural norms, trust, or any other reason. Social and cultural norms play a major role in the lack of participation of girls and women in STEAM education and the ICT sector in Moldova. A whole-of-society approach can only be achieved by constantly scanning for and identifying sources of digital divide and taking nationwide systematic actions to remove inequities and disparities associated with gender, livelihoods, disability and income aspects. At the European Union level, many programs are trying to address inequalities that can be also leveraged by Moldova. The EU4GenderEquality: Reform Help desk program can, for instance, help address gender equality issues.

With regards to digital literacy, it is essential that it should not replace, but rather build upon traditional forms of literacy. The European Commission has developed a Digital Competence Framework for Citizens (DigComp), which provides common understanding across the European Union and beyond of what digital competence is and provides a basis for framing digital skills policy and adapting the digital education programs. Activities under this objective should be aligned with the European Union's Digital Competence Framework (https://joint-research-centre.ec.europa.eu/digcomp/digital-competence-framework_en), which identifies 21 competences across 5 areas:

Figure 6. DigComp conceptual reference model



Source: European Union Digital Competences Framework

Furthermore, the European Union's DESI should be used to measure the progress of Moldova in the digital society, as well as both the efficacy and impact of initiatives in this area.

Achievement of this general objective will be ensured through the following priority directions:

1) **Make digital literacy an integral part of basic literacy education process**. As today's children are exposed to digital technologies from the early years and our society's heavy reliance on such technologies, it is hard to believe that someone can be successful in today's world without strong digital skills in addition to general literacy and skills. By integrating education of digital skills from the start of the educational process, we can ensure that the adults of tomorrow are properly equipped to benefit of and contribute to a prosperous society. This, in turn, requires a sustained effort to upskill the current and future teachers with the necessary competencies, as well as properly equipping the educational institutions of all levels.

2) **Building a proactive educational system and including** that not only teaches digital technologies and computer usage but also leverages digital technologies to teach all other subjects. Integration of digital tools in the education process will maximize its benefits and increase education system resilience for future shocks, such as the need to switch to hybrid or online-only education models

3) **Using data for decision-making, including in the education sector**: Data and integrated digital tools in educational and management processes should be used to drive decisions continuously, on a rolling basis, so that the education system responds to ever-changing needs and priorities and is able to support growth of the ICT sector, the digital economy and, ultimately, of the digital nation. Higher education institutions should proactively seek to develop highly skilled specialists required for the labor market, such as data scientists, artificial intelligence (AI), meta language (ML), Internet of Things (IoT) specialists, user experience/user interface (UI/UX) designers, robotics etc.

4) **Promoting digital skills as core life skills**, treated as a national educational priority at all levels of education (preschool education, primary education, secondary education (cycle I: middle school education, cycle II: high school education), secondary vocational technical education, higher education, continuing adult education). Digital skills should generally be devoid of any negative connotations and social and cultural prejudices, systematically reducing the digital divide between rural and urban areas/young and adult populations and promoting the advantages offered by the technologies in terms of public and private electronic services, offering various stimuli (cashbacks, discounts, reduced delivery time, etc.) for the use of electronic services, etc.

5) **Conducting national digital literacy programs targeting everyone in society**: This would, among other things, explain the benefits and possibilities, but also the risks of the digital space. Such programs should provide a variety of delivery channels and training methods, including formal training, community courses and gamification, but also offering various incentives to stimulate learning of digital skills. Likewise, a digital skills certification framework shall be developed and approved in alignment with DigComp. This would include creation and implementation of parenting skills development programs that take into account the benefits and risks of use ICT, focusing on educating the child about safe online behaviors, solutions and support services available in case of abuse.

6) **Implementation of measures for child protection against online risks**. Supporting professional training of specialists in child protection system by assessing the risk of online sexual abuse, intervention actions and providing assistance according to identified needs. Establishing a coordination mechanism for child protection against online risks, involving representatives of all authorities and public institutions involved in policy making, and creating a working group to plan all actions of promoting children's online safety.

7) **Promote Internet access as a basic human right**: To ensure that every individual has access to quality affordable Internet, the government should undertake a set of measures to make the Internet access a basic human right, alongside healthcare and education, including but not limited to expanding the use of Universal Service and Access Fund (USAF), financing and/or compensating the net cost of fulfilling the obligations to provide the universal service, promoting the shared use of the physical infrastructure by Internet service providers and providers of mobile electronic communications networks, etc.

8) **Involve all sectors in improving digital literacy and skills**: This priority direction includes establishing a national coalition for jobs and digital skills to coordinate and implement national-level digital skills initiatives focused on bridging the gap between the education offering and the needs of the private sector and civil society; training more IT experts; reskilling and

upskilling the labor force; and the development of national digital skills program, various pilot projects, sandboxes, etc.

9) **Adapting human resources policies**, primarily in the public, but also in the private sector: This should be encouraged to leverage ICT in the process of staff hiring, training, and retraining, as jobs increasingly relyon digital skills and abilities. Attention should be paid to reconsidering existing roles and introducing new roles that are essential for driving the digital transformation agendas at all levels (e.g., CIOs, CDOs, etc.). Targeted government programs will stimulate and finance ICT (re)training of workforce in key economic sectors to compensate for the lack of specialized skills. As there will always be difficulties attracting and retaining highly skilled ICT staff in the public sector, policies should be flexible in allowing innovative engagement models, such as outsourcing, extended teams, etc. To implement government programs, they will hire specialists with national or international qualifications, depending on their complexity.

10) **Involvement of local public authorities** in all phases of digital transformation: Local public authorities should be a strong catalyst for grassroot digital transformation activities and are essential for promoting digital skills and opportunities. This, however, requires strengthening the digital skills and capacity of the local public authorities themselves in the first place

11) **Involving citizens in political life using digital technologies**: Developing and promoting citizen engagement platforms such as e-petitions and participatory budgeting, but also introducing electronic decision-making tools such as civic initiatives, collective petitions, referendums and Internet voting, will make sure that people can feel connected to the governance process and will allow them to exercise their civic rights irrespective of their location. This will be increasingly reliant on secure means to identify oneself and transact remotely, hence it is especially important to offer citizens intuitive, affordable, and secure digital identification means.

12) **Leveraging public service delivery offices to assist and instruct people in using digital services**: Switching the focus from offline delivery of public services to assisted delivery of electronic services, whereby people are encouraged to use self-service are aswith the help of civil servants, is a great way to decrease the load of service delivery offices while providing support and giving people confidence in their skills

13) **Involvement of the diaspora in all digital transformation projects**: All the efforts to promote digital skills and services, training programs and initiatives should consider and fully engage the diaspora, eliminating potential engagement obstacles such as the need for physical presence for any activities and leveraging consular networks and communities, which will lead to tighter links between Moldova and its diaspora. Border crossings could be used as diaspora information and promotion points, where advantages and possibilities of digital technologies and services could provide support to representatives of the diaspora in capitalizing on the available digital tools.

Subsection 1. **Results**

As a result of implementation of these priority directions, most citizens will have basic digital skills and the access to internet will be universal. The national education system in 2030 will be able to provide high-quality, inclusive and equitable education to all children/students/adults throughout life, at all levels of education, in various formal/non-formal/informal contexts, will meet the needs of those who learn and of society; it will become more resilient, flexible, providing education without interruption in the ever-changing so-

cial, economic, demographic conditions; will become a real factor in ensuring sustainable, social and economic development by forming quality human resources and will be an essential factor for the development of the knowledge-based society.

The society trusts and widely uses digital solutions to fulfil their needs as well as exercise their civic rights and obligations. The service delivery channels rely on digital platforms and are accessible for everyone, to reduce the digital divides. People are in control of their data and able to securely share it and monitor its use.

Subsection 2. **Indicators**

1. More than 50 per cent of the population has access to their electronic health records
2 More than 70 per cent of citizens have an active digital identity
3 More than 80 per cent of population have basic digital skills
4 The share of STEAM graduates in the total pool of university graduates is at least 16 per cent
5 Moldova's ranking in the Digital Skills Gap Index is in the top 50 countries
6 Minimum broadband Internet access speed is at least 100 Mbps
7 5G coverage represents 25% of population.

Subsection 3. **General objective no. 2. Development of a robust and competitive ICT sector**

There are several objective reasons why digitalization can and should increase inclusion and act as a powerful enabler in a society. Distance from city center, density of population or transport connections are much reduced and access to Internet becomes much more relevant in using public services. Fast and accessible Internet in all parts of the country is not only an enabler, but a precondition for digital development.

More than three out of four citizens of the Republic of Moldova are using the Internet as of 2022, and the number of mobile connections exceeds the total population. 98 per cent of localities have fiberoptic links, but adequate high-speed home access to broadband is still a challenge in specific areas. Constant upgrading of modern ICT infrastructure (broadband, mobile data, public computer access, etc.) in close cooperation with private sector enablers such as electronic communications providers is a cornerstone of access to digital services.

Thus, we will be able to stimulate the growth of the ICT sector and the startup community, which will make an important contribution to digitization and will create strong synergies between the public sector and society.

Achievement of this general objective will be ensured through the following priority directions:

1) **Access to fast and affordable Internet for every citizen of the Republic of Moldova**: Even though 4G and fiber optic coverage of the population is at 98 per cent, the results of the 2021 Annual National Survey revealed that 34 per cent of the unconnected invoke cost as the biggest impediment. Additional measures are needed to ensure 100 per cent broad band (through 4/5G and/or fiberoptic) connectivity, as well as to ensure universal affordability of Internet

2) **Establish a broadband Internet mapping system** to equip the government with a tool to address the digital divide and promote economic development. The broadband mapping systems would be particularly important for determining the availability and accessibility of high-speed Internet in a given area within the country. Such a tool would allow the government, service providers, and other stakeholders in the country to make evidence-based decisions on how to improve broadband in the country and achieve homogeneous digital transformation.

3) **Develop supportive conditions for the ICT sector and start-up community**: While the basic use of computing devices can be considered as modern literacy, ICT specialists with complex professional skills are needed for ICT (and wider technology sector) development. For a vibrant start-up development, early-stagefinancing, and modern start-up-specific regulations such as competitive taxation of stock options are essential.

4) **Consolidate the digital architecture in the public sector**: Although digital services can be developed by different government institutions, there is a clear synergy in the consolidation of their architecture, including software components, advanced and widespread use of cloud services to increase efficiency and security

5) **Joint use and reuse of data and services**: As data is the most important asset in offering electronic services, it is vital to maximize the effective usage and take steps to improve the accuracy of collected data. Once a citizen or business is asked to provide information to the government, it is possible and for more than one public office to use this data, assuming there is a legitimate reason and/or the citizen's consent. By making data available to the private sector, new and innovative services will become available

6) **Increase joint projects between private and state institutions** including the creation of a talent pool of specialists from different spheres of life and with different ICT skills who are invited to contribute to building the digital society / alt: digital government projects

7) **Encourage the use of digital services**: As providing services digitally is more cost-effective for the government and should therefore be preferable, this should be reflected in lower fees for the citizen and/or shorter delivery time due to process optimization.

8) **Promote the digital agenda as a clear, beneficial, and productive investment**: To offer high-quality digital services, a substantial initial investment must be made. Once the service is up and running, the cost per citizen served will be much lower than before

9) **Attract digital businesses and foreign investments to Moldova**: The perspective of becoming a European Union Member State with unlimited access to the common market of almost half a billion people will drastically raise the attractiveness of Moldova as a home for international digital businesses. Highly skilled people, competitive legislation, facilitated tax regime of the virtual IT park and strong infrastructure will all contribute to the attractiveness for foreign direct investment and digital businesses moving to Moldova

10) **Attract qualified IT specialists and investors from the European Union to the virtual IT park**. The full exploitation of the IT mechanism, as well as the remote work abroad of the employees of the virtual IT park residents, establishment and management in the Republic of Moldova of remote IT and innovative businesses.

11) **Promote widespread digital services and e-ID use by the diaspora**: As distances are shorter and communication quicker in the digital space, digitalization can bring the citizens of the Republic of Moldova closer to each other, and the diaspora closer to Moldova.

12) **Study and take over the best practices from the European Union**: Conducting studies on the premise behind the development of emerging technologies, and capitalizing on European Union country initiatives and programs in the field of AI, robotics, blockchain, smart contracts and other emerging technologies, will contribute to their implementation and use and making a qualitative leap of the Republic of Moldova technologically and, respectively, economically.

Subsection 1. **Results**

Digital infrastructure and talents act as critical enablers in making comprehensive digital society a reality. Ensuring access to high-speed internet to all the people of Moldova will lead to higher use of digital services and abolish the service access barriers resulting from geographical distances or rural/urban divide.

More people with ICT skills lead not only to more advanced digital society, but also to an attractive business climate and vibrant startup ecosystem.

Subsection 2. **Indicators**

1) The share of ICT Specialists will be at least 2 per cent of adult population, including gender convergence
2) The share of ICT Companies will be at least 10 per cent of active companies
3) The share of ICT products and services in the total export will be at least 15 per cent.

SECTION 3. General objective no. 3: Creation of an innovative and resilient digital economy

The Republic of Moldova has most of the elements required for the development of the digital economy. In many successful countries, digitization is the direction that ensures the fastest economic growth, efficiency and optimal use of available resources. In medium terms, the focus will be on several important components for economy digitization, supported by the favorable regulatory and policy framework on: the local IT industry and technology start-ups geared towards the international market; competitive ICT infrastructure and services; and e-commerce infrastructure development efforts (diversification of the online payment services market, postal and courier network, digitization of export customs processes, access to international e-commerce platforms, etc.).

In order to reach out the priority directions proposed in this Section, it will be necessary to develop a Digital Economy Development Program in the medium term, as well as to monitor the inclusion of the required actions set in the annual activity plans of the concerned public authorities, aligned with the National Development Strategy 'European Moldova 2030', approved by Law no. 315/2022, MDTS and the Government's Activity Program. Promotion and support of digital innovation in the real sector of the economy changes the way companies manage their processes, integrating digital technology wherever possible within the company and in its relations with authorities, business partners and consumers. Increasing adoption of digital solutions within companies is the tool on which the increase in performance and economic efficiency is based. Digital dynamism in the business envi-

ronment through regulatory measures, monetary or tax stimulation and information shall be encouraged.

For business digitization and its impact on sustainable economic growth, it is essential to grow a dynamic community of local startups in various verticals of the real sector of the economy. For this purpose and to make local startups internationally competitive, there is a need for continuously education and attraction of talents, technological transfer through implementation in production and services of the results of research and development carried out on university platforms, and, not lastly, a favorable regulatory and policy framework for development.

Achievement of this general objective will be ensured through the following priority directions:

1) **Digital transformation of enterprises**: A true digitalization of the economy is based on the digitalization of the entire operational cycle at the level of companies and between companies, internal administrative documents and procedures, financial, commercial records, etc., as well as interaction outside the company, based on these digital resources and tools. Moreover, increasing adoption of digital solutions within companies is the tool on which the increase in performance and economic efficiency is based. Encouraging digital dynamism in the business environment through regulatory measures, monetary, administrative or tax-support and information.

2) **Development and promotion of key digital public services for companies**: Actions in this regard will focus on strengthening the existing legal framework; removing constraints on remote interaction with authorities; improving the applicability of regulations and optimizing existing procedures at the level of institutions that provide public services to companies. In practical terms, these actions address process automation, remote registration of companies, digitization of G2B interaction, facilitation of internal processes in companies and between them, etc. Also, the difficulties in applying the regulations on personal data protection, online availability from a safe and reliable resource of data from the State Register of Legal Entities, cyber security, remote identification, electronic notarization, use of electronic signature and other specific measures to improve the business environment through digitization. The success of these actions will be ensured by regular evaluations, identification and elimination of administrative barriers for the digitalization of companies' interaction with authorities, business partners and final beneficiaries.

3) **Further development of the local IT and BPO industry and assimilation of their products on the domestic market**: existence of innovative industries in the country brings benefits to any economic development pattern, but they are an indispensable element for a modern and digitized economy. Exports of IT services, business process offshoring (BPO) and development of shared service centers (SSCs) will continue their dynamic growth regionally and Moldova is part of this process due to the Moldova IT park. Outsourcing of these services in our country prevails over IT service exports. These revenues will continue their linear growth in the coming years, and the strategic issue consists in assimilating a growing share of these services in the local market, for a positive impact on the development of the local digital economy. Similarly, BPO&SSC are expected to grow at double-digit rates during the same period and Moldova needs to define its public policies in support of these sectors. At the same time, it should be taken into account that our country specializes mainly in provision of IT services with low sophistication. While a number of increasingly complex services are being developed, most projects are relatively low-tech ones. Thus, Moldova's ICT exports have potential to grow as a result of increasing technological complexity, as does most of the country's BPO and SSC export basket.

4) **Stimulation of the market of electronic payment services**: a dynamic digital economy is impossible without various electronic payment instruments, geared towards the local consumer and accessible across borders. This objective can be achieved through open competition and integration of our country into the European/global market of payment services. Cash is the slowest and most expensive means of payment, which stimulates the shadow economy and does not support digital economy development. Availability and use of alternative and accessible online payment methods is a prerequisite for safe and fast financial transactions in a true digital economy.

5) **Reduction of the ICT labor shortage in all sectors of the company**: The number of ICT specialists educated in universities needs to match the market demand for labor force and increasingly diverse qualifications, while the attitude of the authorities concerned concerning the quality and number of specialists trained in the professions of the future shall be changed. Specialized digital training for students of all faculties in the country, in response to the spread of digital technologies in all sectors of the national economy and social life must become a norm. Emphasis shall be placed on upskilling and retraining of employees, particularly from socially vulnerable groups (women, youth, persons with disabilities, etc.) by offering technology induction sessions, training programs, dedicated mentoring, etc.

6) **Promotion of R&D cooperation between subjects of research and innovation, the public and private sector**: Strengthening the links between research and production fosters the results implementation for both universities and the public and private sectors, encouraging open innovations. Mechanisms such as the European Digital Innovation Hubs (EDIHs) can also speed up digital transformation of the national economy. Practical implementation of open science principles and open access to research publications and data will enable a better accessibility and implementation of scientific content. Using the opportunities and integration into the European Open Science Cloud to provide seamless access and reliable re-use of research data to researchers, innovators, companies and citizens through a trusted and distributed data environment and related services. Capitalization on the niches identified by development of the 'Smart Specialization Strategy of the Republic of Moldova until 2030'to support innovation and digital transformation of sectors of the national economy.

7) **Development of a start-up-friendly business environment**: Updating the policy and regulatory framework in close cooperation with the startup community for the development of a friendly business environment, including by establishing sand boxes and innovation hubs for the development of certain verticals, such as FinTech, AgriTech, HealthTech, will make national startups internationally competitive. This direction of actions will positively impact various sectors of the national economy, will encourage identification of niches and partnerships in various industries, for the adoption of digital innovation.

8) **Development of the infrastructure of the digital economy and electronic commerce**: Creation of conditions for proliferation of the digital economy and electronic commerce depends on the development of the necessary infrastructure and removal of regulatory barriers. Reduction of administrative constraints largely depends on elimination of regulatory discrepancies, harmonization with international good practices, facilitation of online payments, access to marketplaces, digital payment services, courier services, etc. The identified actions will serve to achieve this objective, while stimulating competition and varied offer of adjacent services, while transforming e-commerce into an indispensable element of a modern digital economy in the Republic of Moldova and contributing to the absorption of digital innovation in various sectors of the economy.

9) **Development of national platforms and attraction of international e-commerce platforms to the market**: The COVID-19 pandemic has shown a rapid increase in e-commerce in all

of Europe, and it is vital for Moldova to keep up with the clear trend. Online promotion is very affordable in different forms, but in the avalanche of offers, many local manufacturers remain unnoticed. This line of action should support the widest possible integration of local manufacturers in local eCommerce platforms and contribute to an optimal redirection of local consumption towards the online stores of local manufacturers. At the same time, for small manufacturers that are unable to maintain private online stores, digital platforms offer intermediate solutions for sales of products and services. The main objective is a digital product well recognized in the country or a variety of them. Creating functionalities to promote offers for different social interest groups, including through social entrepreneurship is an objective that would support SMEs to enter the economic circuit through digital tools. The part related to the access of international platforms to the Moldovan market is very complex, and depends less on internal factors, primarily on the priorities of investors and the size of the Moldovan market, but it has great potential to facilitate Moldovan eCommerce and eExports. There is a need to identify measures to stimulate and encourage operations in our country by the big digital platforms and co-opt local manufacturers.

10) **Simplification of customs procedures for online exports**: This helps stimulate postal and courier services and increase e-commerce attractiveness. Implementation of digital transport corridors will take into account the high dependence on commercial operations with the EU Member States, in which Moldova can use digital technologies and implement the provisions of Regulation (EU) 2020/1056 of the European Parliament and of the Council of 15 July 2020 on electronic freight transport information.

11) **Transformation of the national electricity network into a smart network and smart-metering**, in order to optimize the behavior of citizens in order to improve energy efficiency in households. Data on electricity consumption obtained through the smart network will be analyzed and used for decision-making to improve policies and the regulatory framework in the energy field.

Subsection 1. **Results**

Digitalization of companies in Moldova and enabling new business models (including through startups that offer disruptive solutions) will lead to more internationally competitive economy, faster economic growth, integration of local manufacturers and service providers in the global market and higher living standard for the entire society.

With digital and e-commerce tools, local companies will become more visible and will have more access to new markets, both regionally and globally, and this will stimulate economic growth, competitiveness of products and services, the level and quality of the life of citizens.

Subsection 2. **Indicators**

1) More than 60 per cent of SMEs reach the basic level of using digital technologies in business management and product and service development
2) More than 40 per cent of companies in Moldova have access to digital platforms for their products and services
3) At least 50 per cent of Moldovan companies use Cloud/AI/Big Data technologies
4) Key digital end-to-end services available for companies, technical possibilities ensured for opening, managing and terminating the business remotely

5) Increasing the export of IT and BPO services by 15% annually and ensuring a degree of local adoption of at least 10 per cent of the digital solutions developed in Moldova

6) Increasing from 6 to 10 per cent the number of students trained in digital technologies in various fields.

SECTION 4. General objective no. 4. Establishment of an efficient, smart and transparent digital state

Digitalization can and should increase the effectiveness, transparency, and security of public services. With widespread use of digital identity and electronic signature, it will be possible to introduce a "digital first" and "secure by design" principles allowing all main public information to be kept and accessed online in a safe and secure manner. There shall not be paper "originals" or duplicating the information held online, as electronic data and records are "originals".

Citizens shall own all the most important data and be able to access logs showing who and when has seen or modified them. Part of digitizing the public services shall be a review of relevant procedures – instead of replicating paper documents online-only, essential information will be gathered and stored.

To ensure accurate information for all state registries and to reduce bureaucracy, a "once-only" principle will be introduced: citizens can only be asked any information once by the state, and other state registries and authorities should ping the original holder of the information.

Services will become proactive, without the need to fill in forms and applications. If a person is entitled to certain rights, they will not be requested by filling in a form, but assigned automatically in most cases.

Achievement of this general objective will be ensured through the following priority directions:

1) **Fully fledged interoperability of platforms and data**: This is a precondition for information exchanges and automation of services, including cross-border services, especially with European Union countries.

2) **Reduction of administrative burden**: Updating and simplifying the regulations is made possible due to the digitalization of systems and reviewing the processes. If a state authority is legally entitled to certain information, pinging a register instead of approaching the subject (citizen or company) is preferred

3) **Development and implementation of the Public Sector e-Transformation Program**: This allows prioritizing and planning digitization activities of services at the proposal of the public authorities/institutions and their connection to the resources available from both the state budget and donors.

4) **Integration of the anti-corruption vision into digital solutions**: By increasing the level of transparency, impartiality and accountability, digital systems will help reduce corruption. Also, processes in the justice sector, such as examination of cases in courts will be digitized.

5) **Migration of state information systems in the government cloud**: This priority direction should be done to increase efficiency and data security. The peak demand for services can vary significantly in time, and using a government cloud can effectively facilitate the required computing power for peak performance and cybersecurity for critical systems and processes.

6) **Creation of digital embassies** to ensure data protection for the most critical registers, especially in the current circumstances.

7) **Citizens have control over their data**: Development of tools which allow people to have better control over and protection of their data, while ensuring data portability, will help people cope with increasing privacy risks and ensure that people will always know and be able to decide who, when and for what purpose their personal data held by the public sector is used.
8) **Automation of state registers**: Enabling interoperability between different state registers will reduce the quantity of data (repeatedly) requested from citizens, improve data accuracy, and increase the volume of information that is pre-filled in online public service forms. Creation of harmonized standards for data entry.
9) **Harmonization and updating of data on natural persons in different registers**, using the person's where applicable, as unique identifier to eliminate information inconsistencies between different sources.
10) **Implementation of AI scenarios in service delivery and data management** by leveraging cloud computing capabilities and existing government data sources for decision making and process orchestration
11) **Strengthening the management capacities of digital transformation projects in the public sector**. Improving the skills of responsible persons from the public sector to implement digital transformation projects will reduce the risks related to the implementation of these projects, and building the skills of employees from the public sector to use data solutions of macro-level analysis can lead to a more accurate understanding of situations and informed decisions
12) **Implementation of automatic decision-making processes in the public sector**. Machine learning makes it possible to predict and automate services thus leading to greater beneficiary's satisfaction and higher efficiency of spending in the public sector.
13) **Proactive, simple and intuitive digital services**: Some of the public services can be offered to citizens without the need to request them explicitly, either as a result of certain life events or in addition to other public services. Such life events can vary from the birth of an only child (rights to allowances, registration of name and address), going through the automatic renewal of documents, to automatic issuance of new documents, when the surname changes as a result of marriage.
14) **Increasing the use of electronic services**: Promoting and improving access to public services through digital tools. This means that beneficiaries can access services they need online in a simple and easy-to-understand way.

Subsection 1. **Results**

Faster and more comfortable access to public services for citizens, automation of decision-making for the government and more transparent and efficient public sector for the whole society.

Subsection 2. **Indicators**

1) Moldova ranks in top fifty countries in the E-Government Development Index
2) 100 per cent of key public services are available online
3) The share of online transactions represents at least 70 per cent of the total public services transactions
4) The share of the use of digital services in judicial processes will be at least 30% of the total number of cases
5) 100 per cent of documents and data in the public sector is exchanged electronically.

SECTION 5. General objective no. 5. Building an accessible, secure and inclusive digital environment

A secure ICT and digital environment are key for all citizens of Moldova to enjoy the full benefits of a secure and resilient cyberspace and are necessary elements for a successful and coherent digital transformation of the Republic of Moldova.

At the national level, cybersecurity is a shared responsibility that requires coordinated actions on the part of government authorities, the private sector and civil society. Considering the global nature of cyber threats, the Republic of Moldova is to implement and maintain a national cybersecurity management system, which will ensure an institutional framework and effective cooperation on cyber security issues and crisis management in the field; establishing norms and international cooperation; resilience of digital infrastructures of critical service providers; adequate analysis and monitoring of the situation and evolution of cyber threats; cooperation with private sector solution providers in promoting cyber security; strengthening capacities and promoting education and professional training in the field of cyber security; harmonization of legislation and effective strategies to combat cybercrime.

The recent adoption of Law no. 48/2023 on cyber security, which transposes the regulatory framework of the European Union, is an important step towards creation of the national cyber security management system. For the successful implementation of the provisions of the law, the necessary resources are to be allocated by the state, as well as attracted from development partners, especially for the building of technological capacities and human resources in the field.

Achievement of this general objective will be ensured through the following priority directions:

1) **Establishment of a competent cybersecurity authority at the national level** with single point of contact functions, cyber security incident response team (CSIRT), as well as identification, monitoring and supervision, operational coordination of crisis situations, cooperation and interaction at both national and international level.

2) **Implementation of the general strategic and operational framework for coordination and cooperation** between the public and private sectors in the field of cyber security (the National Cyber Incident and Crisis Response Plan, the National Cyber Security Strategy, the Cyber Security Coordinating Council).

3) **Implementation of cyber security measures and mandatory mechanism of significant cyber incident reporting** by entities whose services are critical for the functioning of the economy and society, as well as the possibility of voluntary notification of cyber incidents.

4) **Strengthening the resilience of the digital infrastructures of critical service providers** by carrying out cyber security exercises at national and sectoral level; development and strengthening of technological capacities and human resources in the field; adjustment of the sectoral regulatory framework; promoting good practices and cyber security culture.

5) **Ensuring the application of cyber security requirements (standards) in the development and/or procurement of software and hardware** used in provision of critical services, including new generation electronic communications services, as well as supply chain security, so that the products are secured by default.

6) **Building cyber security skills of human resources**. Professional training of civil servants, as well as of any other stakeholders from the public and private sector, through training programs, professional improvement training, cyber security training exercises; development of research and innovation programs in the field of cyber security, including for the protection of public and private critical infrastructure, protection of personal data, security in the use of artificial intelligence, as well as innovative solutions for the prevention, detection and response to cyber threats; development of security guidelines and best practices for accelerated deployment of standardized cyber security solutions.

Subsection 1. **Results**

Following the actions in the proposed priority directions, an increase in the level of cyber resilience of the key entities in the Republic of Moldova will be ensured, thus, they will be managed much more efficiently and transparently, and also cyber incidents, possible material and reputational damages associated with them will be avoided. At the same time, a multidisciplinary approach will be implemented, and all stakeholders, including the Government, the private sector and civil society, will assume their joint responsibility and commitment to support and cooperate in ensuring cyber security

Subsection 2. **Indicators**

1) The Republic of Moldova will rank among the top 50 countries according to the ITU Global Cyber Security Index (https://www.itu.int/epublications/publication/D-STR-GCI.01-2021-HTM-E)
2) Starting with 01.01.2025, the national CERT and the Cyber Security Coordinating Council operate
3) At least 90% of civil servants have skills in cyber security
4) The National Cyber Incident Response Plan and the National Cyber Security Strategy approved
5) Service providers identified and monitored under Law no. 48/2023 on cyber security.

SECTION 6. General objective no. 6. Strengthening Moldova's image as a digital nation

As an official candidate country to join the European Union, Moldova should make it a top priority to comply with all relevant European Union standards and regulations and to become an integral part of the Digital Single Market. This means that all the digital services created in Moldova should be available for use by citizens of other European Union Member States and that cross-border electronic services will ensure better mobility of citizens of the Republic of Moldova within the European Union.

The recently declared common digital space between the Republic of Moldova and Romania is, therefore, a great opportunity to prove that the existing digital platforms and services are reusable, built on open standards and quickly adaptable to the needs of Romania. At the same time, given the European Union membership status of Romania, the common digital space will contribute to development of solutions that are compatible with European Union standards and regulations, facilitating seamless cross-border interactions between Moldovan and European Union citizens and businesses.

As a country that has set digital transformation as a strategic priority and that has been actively implementing state-of-the-art digital transformation projects for the past decade, Moldova will actively promote all the best practices and its digital platforms and services internationally. This will be a coordinated effort by both the public and the private sector, resulting in global recognition of Moldova as an advanced digital nation.

Achievement of this general objective will be ensured through the following priority directions:

1) **Interconnection of data and services internationally**: The world is increasingly connected, and this should result in an increased number of secure cross-border services and data exchanges, especially with the European Union, including through integration in the common European data space. Citizens of the Republic of Moldova should have the mobility to travel, study, work and retire easily abroad, while having the means to provide their host countries with required data and documents to benefit of all the conditions offered by international and bilateral agreements

2) **Creation of cross-border electronic services**: Similarly, using its advanced digital infrastructure, Moldova should strive to simplify the interaction with residents of other countries, be it for the purpose of travel, study, doing, business, etc. Modern applications and contexts should allow for seamless interaction with national or cross-border electronic services. Again, leveraging the common digital space with Romania could result in incredible opportunities

3) **Mutually recognition of electronic signatures and identities**: As in the case with digital identity at the national level, having the ability to securely identify and transact remotely in a different country is essential to ensure cross-border services and allow data exchanges based on individual consent. This will allow seamlessly rolling out e-residency programs that can, then, extend to include citizens of countries that are not part of mutual e-signature recognition but who would like to benefit of Moldova's business and social environments

4) **Participation of in international networks** with countries that share the same values, with a strong focus on the European Union Digital Agenda and gradual integration into all major digital projects: This will ensure that Moldova leverages its digital transformation experience and is able to assist other countries in their digital transformation efforts, while continuously learning from international best practices and improving its national digital infrastructure. One such example of cooperation could be identifying and implementing principles of fair taxation of digital economy, which is proving to be a major challenge for most Governments

5) **Promotion of Moldova internationally as an advanced digital country**: Moldova has not always been showcased internationally in favorable contexts. Changing its image from being one of the poorest countries in Europe with a heavy post-Soviet legacy to being a modern, forward-looking, digital innovator will require sustained efforts from all major stakeholder groups. The results will be worth the effort, offering Moldova multiple opportunities for accelerated socioeconomic development

6) **Promotion of the Moldovan ICT sector internationally as a reliable partner**: The journey of the Moldovan ICT sector from insignificance to contributing over 7 per cent of national GDP has been remarkable. Moldovan ICT companies offer top skills at competitive prices and have earned a reputation for delivering high-quality services. The Moldova IT Park has managed to attract many international companies, offering them good incentives. Future efforts should be directed to increasing the ICT sector and its attractiveness for current and future students, as well as for international companies. Another important direction would be to reorient Moldova from a pure service outsourcing destination to a country that produces high-quality ICT products that retain most added value in-country

Subsection 1. **Results**

Moldova will become fully integrated in European Union's digital space and Digital Single Market. The digital solutions, platforms and services developed by the Government will be interoperable and able to accommodate cross-border scenarios. European Union citizens will be able to benefit of Moldovan e-services using the digital identities issued by their countries, while citizens of the Republic of Moldova will be able to use their digital identities in the European Union cyber space.

The country will become an attraction center for top ICT companies due to local talent and favorable conditions. Moldovan ICT products and brands will become recognizable and appreciated in the world.

Subsection 2. **Indicators**

1) E-signatures are mutually recognized between Moldova and the European Union countries
2) At least 30 per cent of public e-services are available in cross-border scenarios
3) The first start-ups from the Republic of Moldova reach the status of unicorns and at least 10 Moldovan products reach a valuation of US $100M

CHAPTER V

GOVERNANCE MECHANISMS FOR STRATEGY IMPLEMENTATION

Digital transformation is less about technology adoption and more about changing the operating mode and business processes of public and private entities in a way that leverages technology to enhance the user experience, as well as the efficiency and sustainability of organizations themselves.

The complexity of implementing a national-scale digital transformation resides in the fact that it needs to span the entire public and private sectors as well as civil society and reach all organizational units, rather than being siloed and implemented as a typical program. It requires more coordination when making decisions and conducting activities than the usual way of doing business. In the government's case, many digital transformation reforms, projects, and activities need to engage the government in its entirety, with all ministries and agencies rather than individual organizations.

A well-governed digital strategy must be able to satisfy different stakeholders and be flexible enough to accommodate multiple types of initiatives, while ensuring sufficient rigidity to achieve general goals. While traditional governance prioritizes compliance and risk mitigation, the approach for digital transformation should prioritize enablement over control – in other words, remove blockages and impediments to progress through rapid decision-making. (https://www.kommunikationsraum.at/wp-content/uploads/2021/04/Going-Digital_Howtoembracechange.pdf).

Governance mechanisms implemented around digital transformation initiatives show that three mechanisms – shared digital units, organization-level committees and new digital roles – are likely to support the digital transformation and common across organizations for successful implementation of digital transformation strategies. (https://www.capgemini.com/wp-content/uploads/2017/07/Governance__A_Central_Component_of_Successful_Digital_Transformation.pdf).

Therefore, based on the best international practices, sections 5.1-5.4 describe the governance structure designed to support implementation of MDTS should include the following mechanisms.

SECTION 1. Factors of successful strategy implementation

The organizational, human and digital foundation elements, in addition to vision and funding, are all required to convert a vision into tangible results.

1) **Leadership commitment**: According to recent research, less than one third of all digital transformation efforts succeed globally, and the right, digital-savvy leaders need to be in place to improve their chances of success.

2) **Implementation capacity and change management**: The experience of many development partners is that significant budgets allotted for digital transformation programs are not absorbed and benefits are not used due to lack of implementation capacity of public counter-

parts. Project ownership, project and change management and institutional implementation capacity, including coordination within and among institutions, were identified as key skills that need to be developed within government institutions, as they are not functions that can be delegated or outsourced externally.

3) **Digital competencies and skills in all spheres**: These are needed to leverage and reap the benefits of digital transformation initiatives and ensure proper return on investment.

4) **Reduced digital divide**: Supply and demand in digital space need to be better balanced to create the preconditions for digital factors to include more of those who are currently excluded

5) **Appropriate funding of digital transformation programs**: Investments in digital transformation usually pay off quickly and bring multiple long-term benefits, so it is imperative to allocate the necessary resources for interventions in this area. For example, in case of e-services, in addition to improving the user experience and accessibility of the service, the number of personnel members involved in provision of the service is usually reduced, and public data management is improved, enabling better informed decisions.

6) **Enabling legal and regulatory environment**: A proper legal and regulatory framework will facilitate implementation of digital transformation activities, and so the existing framework should be continuously adapted for this purpose. The MDTS does not intend to identify all necessary changes required for its implementation; the important principle here is to make sure that they are aligned to European Union legislation and standards, promote electronic services and the use of technology in all sectors, and eliminates barriers for further development.

7) **Unified service design, delivery and monitoring mechanisms** need to be in place to ensure that benefits of digital transformation can be reaped in all sectors.

8) **Digital identification** is one of the most fundamental building blocks for digital transformations. Without the ability to securely identify oneself and transact online, no complex remote interactions are possible and any further digital transformation efforts will be flawed and incomplete.

9) **Interoperability and the once-only principle**: The seare essential to provide seamless, accessible, human-centric electronic services. For this, standardization, and unified principles for development of electronic services, as well as data reuse and cross-use across various data sources and sectors, are important aspects to consider

10) **Digitization of data from the public sector**: This needs to be complete, including (first and foremost) core records such as civil status, cadaster, criminal records, etc.

11) **Digital-by-default principle**: This needs to be put in practice everywhere. Besides recognizing digital sources as the single sources of truth in favor of paper archives, the next step is to dispose of unnecessary electronic documents in favor of digital records and data. Digital channels should be used as primary service delivery channels

12) **Trust in digital services**: This is one of the less visible essential elements for the adoption of digital technologies and e-services, it involves issues related to cyber security (including the implementation of the secure-by-design principle), control over own data, transparency and existence of feedback mechanisms.

13) **Countrywide, high-speed, affordable, and reliable connectivity**: This is another foundational element that ensures sufficient traction on the adoption side, allowing for widespread use of technologies and electronic services.

14) **Use of robust, and forward-looking technologies**: When digital transformation is happening in a whole society and affects all sectors of the national economy, choosing the rights technologies that are fit for purpose, that have been tested in mission-critical enterprise/government environments and that are embraced by wide communities of practice is another precondition for the sustainable development of digital infrastructures and services.

SECTION 2. Leadership and oversight

The leadership and oversight mechanism will ensure proper coordination of strategy implementation and will rely upon the **Ministry of Economic Development and Digitization, together with the Electronic Government Agency** that will ensure a complete and transparent picture of the entire portfolio of digital initiatives with in the strategy.

MEDD is the public authority responsible for developing policies in the field of the digital economy, the ICT sector, cyber security and digitization of public services.

EGA is responsible for the implementation of policies on digital transformation, modernization of government services and the use of ICT and technological innovations within the government institutions. Given the importance of the implementation capacity, the Electronic Government Agency will, therefore, be strengthened with additional capabilities in the form of two organizational structures - a Strategic Delivery Unit.

The Strategic Delivery Unit will be responsible for implementing core initiatives under the strategy that are essential to ensure its successful implementation. As implementation capacity is a continuous struggle for the public sector, the Strategic Delivery Unit will help alleviate this issue and provide strong management and implementation capabilities. Proximity to the MEDD and the EGA will ensure short and efficient escalation paths and mechanisms to quickly remove operational roadblocks.

SECTION 3. Coordination

The main body coordinating the implementation of the strategy is proposed to be the **National Digitization Transformation Council (NDTC)**.

The Council will be created by Government Order and chaired by the Deputy Prime Minister, the Minister of Economic Development and Digitization and will include members from the public and private sectors, the ICT industry, civil society.

If necessary, working groups can be established under the NDTC, responsible for the priority directions of MDTS.

The NDTC will coordinate the MDTS implementation and approved programs to succeed in the priority directions.

The **NDTC Secretariat** will be provided by the MEDD and will include the necessary support for the organization of the Council's activity and the dedicated working groups.

The **Economic Council under the Prime Minister of Moldova** is an advisory body tasked with ensuring a constant dialogue between the business community, the donor community and the Government to develop a sound, non-discriminatory, transparent and investment-friendly socioeconomic and business environment, identifying constraints for

business and developing draft legislative measures to facilitate innovation in the business-to-government and business-to-consumer relationships in terms of modernization through digitalization. The Economic Council under the Prime Minister of Moldova will oversee and contribute to the implementation of activities under the General Objective no. 2.: Development of a robust and competitive ICT sector and the General Objective no. 3: Creation of an innovative and resilient digital economy.

Cyber Security Coordinating Council. To ensure the strategic coordination function in the field of cyber security, the Government will establish the mode of organization and operation of the Cyber Security Coordinating Council, a collective body without legal personality, whose basic function will be promotion and coordination, at strategic and operational level, of cyber security policies.

SECTION 4. Shared digital units

Implementation of the MDTS at the level of implementing entities in the public sector will be supported and facilitated by the shared digital units (EGA, ITCSS, PSA, ALRC), which will:

1) Develop and operate shared platforms and services, used by multiple organizations (e.g., hybrid cloud, interoperability, digital identity, front-office).
2) Help implement the prerequisites for digital-by-default, secure-by-design, once-only (*digital by default*, *secure by design*, *one time*), and unified service design, delivery, and monitoring principles.
3) Offer capacity support to other public sector implementation entities in the fields of project management, business analysis, public service modernization and provision, cybersecurity, etc.
4) Identify and develop new digital skills, best practices, and innovations to support implementation of the strategy.
5) Provide the necessary metadata publishing support and solutions for digital spatial data and network services.

The **Public Services Agency**, besides record-keeping and operating the mission-critical registers and databases for civil records, population, cadaster, business, and licensing, is the largest service provider in the country and operates a network of multifunctional service delivery centers that will play an essential role in enhancing service delivery, raising awareness with regards to the use of digital technologies and promoting the development of digital skills.

The **Information Technology and Cybersecurity Service** ensures the administration, maintenance and development of the information technology infrastructure, the telecommunications system of public administration authorities and management of the unique Government's public key infrastructure; implements the state's policy in the field of cybersecurity; and operates the Government Computer Emergency Response Team.

SECTION 5. Support and advisory

As part of the implementation of the MDTS and the priority direction programs, an essential role will be attraction of advisory and technical assistance from development partners and partners from the private and associative sector. In this sense, based on the needs

identified by the entities responsible for implementing the priority directions, specialized associations, specialized private companies, and most importantly the support of development partners active in policies supporting digital transformation, digital innovation, the economy and the digital society will be called upon.

The representatives of the development partners, professional associations and private companies participating in the projects that will contribute to the achievement of the MDTS priority directions will be able to be included in the NDTC membership and the dedicated working groups.

SECTION 6. Sectoral implementation and funding

The responsibility for implementing this strategy lies with all the ministries, agencies, and other central administration authorities, as well as the local public administration authorities. The responsibilities will be detailed within the programs that will be approved by the Government for the achievement of the MDTS priority directions.

For a better collaboration with the MEDD, the EGA and the NDTC, each ministry, agency and central public administration authority will appoint a Digital Transformation Coordinator (DTC), at the level of state secretary or deputy director. The DTC is responsible for coordination of digital transformation activities for that institution. The DTC will be assisted by the EGA to create internal capacity required to drive the digital transformation agenda and ensure a balance between internal capacity and outsourcing. Appropriate funding mechanisms will be identified using own institutional budgets, support from development partners and other legal sources.

The implementation of this strategy will also involve the regulatory authorities according to their competence, as well as academia, businesses, civil society, and citizens.

The financing of the activities for the implementation of the MDTS and the priority direction programs will be ensured within the limits of the allocations approved in the state budget for the implementing authorities/institutions, and also from the support granted by the development partners.

Activities with financial impact for the maintenance of existing equipment, platforms and services, as well as for the development of new infrastructures, will be planned in the process of drafting the Medium-Term Budget Framework (MTBF) and the State Budget Law for the year concerned.

Multi-Donor Trust Funds (MDTF) and programmatic budgeting could be managed by the development partners to finance specific activities under MDTS. The MEDD with the EGA will coordinate the priority directions that will be included in the country programs of the development partners and respective MDTFs and programs, to ensure adequate synergy between various implementation programs and funding streams.

Another potential source of financing could be European Union's Digital Europe Program, the negotiations for adhesion to which have already been initiated with the European Commission.

At the same time, the resources offered by other projects/initiatives of the European Union for the countries of the Eastern Partnership will be called upon and used.

CHAPTER VI

MONITORING AND REPORTING PROCEDURES

SECTION 1. Expected impact

The MDTS underlines the most important aspects of the digital development of the state, society, and economy. The strategy defines a set of priority directions, which will help achieve the vision and objectives of the strategy, as well as bring multiple benefits for all.

1) Digital Moldova is built using the whole-of-society principle, benefiting everyone, and leaving no one behind.
2) Electronic services focus on people's need and are inclusive, proactive, accessible, secure, and easy to use. Services are paperless, accessible from anywhere, using delivery channels that accommodate everyone, thus creating all prerequisites for turning Moldova into an invisible digital state.
3) The society enjoys an efficient, predictable, secure, cohesive and innovative public administration at all levels, which relies on widespread use of digital technologies to perform its functions.
4) The economy is knowledge-based, agile and innovative, the business environment widely uses digital solutions, and the business support framework is fully functional. Most financial transactions and operations are digital.
5) Administrative and personal data are reliable and secure; cyberspace is well protected; the rights of everybody in digital space, including personal data, are assured.
6) Moldova is a recognized digital state, digital development happens in all sectors of life and in every place of the country, and the digital gap is low from all perspectives.
7) Most people and institutions in Moldova are equipped with digital skills and enjoy high-quality connectivity and tools to make full use of the benefits and opportunities of a digital society.
8) The society is highly skilled and fluent in digital terms; adults' learning and education for digital skill building and capitalization on informational technologies are an important aspect.
9) Digital transformation is an inalienable part of all national development programs and plans, irrespective of their sources of financing, and is intended to support the general objectives of the Moldova 2030National Development Strategy.
10) The digital solutions turn Moldova in an environmentally friendly country.

SECTION 2. Main success, monitoring and evaluation indicators

Digital transformation is a social learning process, sustained over time, involving diverse stakeholders. Its ultimate objective is to harness the global digital revolution to meet a country's specific socioeconomic priorities. It is driven by vision, leadership, innovation, learning and partnerships among the Government, business, academia and civil society. (https://blogs.worldbank.org/digital-development/how-can-developing-countries-make-most-digital-revolution).

The monitoring process of strategy implementation will be organized through a structured, transparent, and shared set of monitoring and evaluation indicators of planned actions, through which the degree of strategy implementation is measured and through which the level of achievement of the objectives is established.

The results indicators will assess the immediate and medium-term effects produced by the implementation of MDTS on different target groups.

Framework of result indicators

#	General objectives	Indicators	Benchmarks (2022)	Targets for 2026	Targets for 2030	Responsible authorities/ institutions
1.	**Development of a digital society**	**eHealth:**				Ministry of Health,
		Share of citizens with access to medical records	0%	>25%	> 50%	e-Governance Agency
		Digital identity:	9%			
		Share of citizens who have an active digital identity		30%	>70%	Public Services Agency
		Basic digital skills:				National Bureau of Statistics
		Share of the population with basic digital skills	N/A	>20%	>80%	
		STEAM Graduates:				Ministry of Education and Research, National Bureau of Statistics
		Share of STEAM graduates in total university graduates:	13,7%	>14%	16%	
		Digital Skills Gap Index:	Position 73 of 134	Position 69	Position 50	https://dsgi.wiley.com/global-rankings/
		Connectivity:				National Regulatory Agency for Electronic Communications and Information Technology
		Share of the population with 100 Mbps connectivity	12%	50%	75%	
		Share of population with 5G coverage	0%	5%	25%	
		Number of accesses to the INDS Geoportal and the EU INSPIRE Geoportal	N/A	50%	75%	Agency for Land Relations and Cadastre https://geoportalinds.gov.md
2.	**Development of a robust and competitive ICT sector**	**ICT specialists:**				National Bureau of Statistics
		Share of ICT specialists in the adult population, including gender convergence	2,7%	2%	>4%	
		Share of the gross added value of the 'Information and communications' activity in GDP:	6%	7,5%	>10%	National Bureau of Statistics
		Share of ICT services in total service exports	10%	14%	>15%	National Bureau of Statistics

#	General objectives	Indicators	Bench-marks (2022)	Targets for 2026	Targets for 2030	Responsible authorities/ institutions
3.	**Creation of an innovative and resilient digital economy**	**Use of digital technologies:** Share of enterprises purchasing cloud computing services; Share of enterprises analyzing big data from any data source; Share of enterprises using artificial intelligence technologies.	n/a	20%	>50%	National Bureau of Statistics, e-Governance Agency
		Digital intensity: Share of enterprises with a basic level of digital intensity (at least 4 technologies used out of 12 selected)	n/a	20%	> 60%	National Bureau of Statistics
		Electronic commerce: Share of enterprises that carry out web sales of goods and services	0,9%	15%	> 40 %	National Bureau of Statistics
		Share of essential online digital services available to companies (opening, management, closing a remote business)		75%	100%	Public Services Agency
		Electronic invoice: Share of companies that use electronic invoice			>50%	State Tax Service
4.	**Establishment of an efficient, smart and transparent digital state**	**EGDI e-Government Development Index**	Position 79	Position 70	Position 50	UN survey on e-government
		Share of public services available online	40%	80%	100%	e-Governance Agency, public service providers
		Degree of use of electronic public services intended for natural persons: Share of public services accessed online by natural persons from the total provided public services, %	n/a	40%	65%	e-Governance Agency, public service providers
		Degree of use of electronic public services intended for legal entities: Share of public services accessed online by legal entities from the total of provided public services, %	n/a	60%	85%	e-Governance Agency, public service providers
		Exchange of documents and data in the public sector: Share of the exchange of documents and data in electronic format of the shared total data	n/a	60%	100%	State Chancellery, e-Governance Agency

#	General objectives	Indicators	Bench-marks (2022)	Targets for 2026	Targets for 2030	Responsible authorities/ institutions
5.	**Building an accessible, secure and inclusive digital environment**	**Global Cyber Security Index**	Position 63	Position 59	Position 50	International Telecommunication Union
		Functional national CERT:	Nu	Da	Da	Ministry of Economic Development and Digitization
		Share of civil servants trained in cyber security	n/a	60%	90%	State Chancellery, Technical University of Moldova
		Share of Government IT Systems placed in MCloud	n/a	75%	95%	Information Technology and Cyber Security Service, e-Governance Agency
6.	**Strengthening Moldova's image as a digital nation**	**Countries that have adopted the experience and digital solutions of the Republic of Moldova**	n/a	1	3	e-Governance Agency
		Number of foreign delegations visiting the Republic of Moldova and to whom the digital experience of the Republic of Moldova was presented	n/a	50	100	e-Governance Agency
		Share of EU countries in which Electronic Signatures are mutually recognized with the Republic of Moldova	0%	50%	100%	Information Technology and Cyber Security Service
		Share of electronic public services available in cross-border scenarios from the total of electronic public services	0%	>10	>25%	e-Governance Agency
		The first startups from the Republic of Moldova reach the unicorn status	0	0	2	Ministry of Economic Development and Digitization
		Number of Moldovan products valued at a minimum of 100 million US dollars	0	2	10	Ministry of Economic Development and Digitization

SECTION 3. Implementation risks

Identifying, evaluating risks, anddeciding on the measures to respond to, mitigate or eliminate them are important aspects of the strategy implementation process. The relevant authorities involved will be the risk owners and shall have risk management plans in place.

The following table shows potential identified implementation risks and their possible mitigation measures.

Risks	Effects	Level of probability and impact	Prevention or mitigation measures
STRATEGY			
Political instability and/ or lack of agreement at the level of political forces in the legislature on the country's digital transformation and/ or legislative and budgetary priorities	Delay or cancellation of the implementation of some objectives	Medium	1. Awareness, discussions, continuously informing political parties/leaders on the benefits of digital transformation as an investment 2.Political leaders should be involved from the beginning 3. Governance structures of MDTS will include funding as a priority activity.
Adverse developments related to international situation, in public health, environment, and technologies (war in region, unforeseen epidemics of infectious diseases; natural disasters or major accidental pollution; industrial and computer accidents, etc.) that may affect the achievement of general MDTS objectives	Delay or cancellation of general objectives	Medium to high	Monitor the international situation and prepare/ update risk management plan/risk register; consult with development partners;
Lack of a unitary and systemic approach (coordinated) in the process of elaborating long-term strategic planning documents (for the implementation of the National Development Strategy "European Moldova 2030", Strategy for an Inclusive, Sustainable, and Digital Economy until 2030 for public policy areas within the competence of the government, relevant to achieving general MDTS objectives	Low performance in achieving general MDTS objectives due to incomplete and/or unclear transposition (takeover) in subsequent planning documents (medium-term programs, etc.)	Medium	Analyze existing problems and amend relevant documents/government decisions

Overestimated strategic targets (in terms of time and value)		Low	Continuous monitoring and adjustment of targets to the actual situation
OPERATIONS			
Limited institutional capacities for the operationalization of MDTS (elaboration and implementation of subsequent documents: annual, medium-term public sectoral policy planning programs, budgetary programs, etc.)	Deficiencies in the use of MTBF mechanisms and, as a result, a lack of financial resources for general objectives fulfillment	High	1.Ensure institutional capacities (including competencies and procedures) for long-term budgetary planning and strengthen the level of correlation between sectoral and financial (budgetary) strategic planning; 2.Ensure the participation of representatives from all interested parties in the coordination structure and the prioritization of planned actions to minimize the risks
Limitations and deficiencies of relevance and credibility (accuracy) of the system (including statistics) of collecting, analyzing, and presenting the data necessary for the process	Lack of comparable data for indicators for planning and international comparability	High	Integrate European Union standards and methodologies on collecting data
Inconsistencies between the resource allocation mechanism (budget planning and public procurement) and the process (normal cycle) of implementing projects for the execution of works and services	Delays in achieving the planned objectives, affecting deadlines and quality	Medium	Exclude the inconsistency between resource allocation mechanism and the process of project implementation

Objectives	Indicators (targets for 2030)		Notes/Data sources
Weaknesses in the system of employment and motivation of human resources in the public service, especially within the implementing agencies, which generate the lack and/or departure of qualified personnel, as well as the reduction of institutional capacities (https://www2.deloitte.com/bd/en/pages/public-sector/articles/digital-government-transformation.html)	Delays in attaining the targeted objectives	High	Strengthen the motivation system for obtaining high qualifications and performance in the public service, including the adequacy of the salary level at the level of the labor market; reconsider the employment and motivation system of the staff within the implementing agencies
Lack or insufficiency of complex digital competences of employees of companies;	Failure to achieve the set objectives	High	Improve the digital skills of employees Organize study visits to similar companies in the country and in other countries
Institutional resistance to digital transformation reforms	Resistance to change may affect institution digital transformation	High	Allocate budgetary resources for programs to increase the skills of civil servants (training, internships, good practices etc.), especially in terms of change management
Operational performance affected by the imperfection of the risk management system	Delay or lack of implementation of planned actions due to an inadequate risk management system	Medium	Strengthen the normative and institutional framework (including methodologies, procedures, and competencies) to ensure the effective implementation of performance and risk management systems within public institutions responsible for achieving the objectives of policy documents

TECHNOLOGY			
Potential for losses due to technology failures or obsolete technologies	Technology-related Risks have an impact on systems, people, and processes; priority risk areas may include	High	Integrate European Union legislation, regulations, and standards on new technology Adopt solutions that can securely share data insights and can flexibly scale up and down to respond to changing conditions and demand
Low level of new technology acceptance and implementation Underdeveloped and obsolete systems (non-interoperable IT platforms and tools)	Increase of the digital gap with European Union and Eastern Partnership countries in implementation of new technologies	High	Comply with good practices in advanced regions and countries; strengthen the public-private partnership in using technologies Periodically replace obsolete technologies
FORENSICS, CYBER, RESILIENCE, DATA LEAKAGE			
Weaknesses in the system of security (preservation) of critical data (information) and protection against unauthorized access and computer viruses	IT system unauthorized penetration may affect the functionality of state information systems	High	Audit existing IT systems on security and protection Design a risk-based architecture for data Protect and manage IT systems
Hybrid war in the region and cybersecurity risks	Intensive cyberattacks may affect the functionality of state information systems, registers, and other basic IT systems	High	1.Operationalize the Cyber Coordinating Council to ensure information security 2.Cooperate with European Union entities to address risks
THIRD PARTIES			
Reduction or delay of budget allocations for financing the priority directions of the MDTS, caused by unforeseen expenses that may arise due to wars and crises, etc.	Delays in the implementation of general objectives	High	Review the objectives Request for EU funds for support infrastructure

SECTION 4. Monitoring, evaluation and reporting mechanisms

The strategy sets the basic indicators for evaluating the level of achievement of its objectives. Considering the status of candidate country for accession to the European Union (EU) of Moldova, the framework for monitoring and evaluating the digital transformation will be based on the DESI indicators and on the Methodological Reference Document on e-Government 2020-2023 (https://ec.europa.e-u/newsroom/dae/redirection/document/88734), which measures performance of the EU Member States.

According to Government Decision no. 386/2020, during the MDTS implementation, the following monitoring and evaluation procedures will be carried out.

1) The **annual monitoring** will be carried out by the Ministry of Economic Development and Digitization, based on the information provided by the sectoral implementing authorities/ institutions. Annual progress reports on the MDTS implementation will be drawn up based on (i) subsequent public policy documents (medium-term programs), (ii) planning documents of public authorities responsible for the MDTS implementation, and (iii) collected monitoring indicators. The annual reports will include a presentation of the degree of implementation of the actions/programs derived from the strategy; of the degree of compliance of the actions undertaken with the plan; an identification of the causes of delays and changes to improve the program implementation process; and an analysis of the dynamics of the evaluation indicators included in MDTS, including those related to risk management. Recommendations will be made to improve implementation of the programs derived from the strategy.

2) In 2026, upon the expiry of the intermediate stage of MDTS implementation, an **intermediate evaluation report** will be drawn up. Criteria such as relevance, effectiveness, efficiency, sustainability and impact of strategy implementation will be considered. When evaluating the implementation of MDTS, progress in achieving intermediate indicators will be taken into account and an adjustment of activities and final indicators will be proposed, depending on progress and trends at both national and international levels.

3) Upon completion of MDTS implementation, a **final evaluation report** will be developed with input from key implementation stakeholders. The annual reports will be published on the MDTS website by 15 April of the following year. The intermediate and final evaluation reports will be made public by 1 September of the year following the evaluation. The final evaluation will involve both key partners and key constituents of the specialized central body to ensure an objective and comprehensive process.

In addition to formal reporting mechanisms, given the rapid pace of development of the digital environment, the EGA will develop and make public a *real-time reporting dashboard* to provide constant progress indicators and possibilities for intervention and corrective activities in useful time.

The MDTS and the EGA will continuously ensure wide coverage of implementation of this strategy, tailored to different categories of public, including general public, business environment and development partners.